QUALIFICATION STANDARD
AND PLAN (QSP)

Naval Combat Systems Assistant Head of Department (NCS ENG AHOD)
NQual (AISF)

MITE ID
OJT – 114808

Regular Force: Naval Combat Systems Engineer 00344
Primary Reserve: Naval Combat Systems Engineer 00344
Training Authority (TA): Commander Naval Personnel and Training Group (CNPTG)
Curriculum Control Authority (CCA): CO HMCS Venture

MARPAC RDIMS # 1340085
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RECORD OF AMENDMENTS

1. The following versions of this QSP have been approved for use in support of the conduct of training. Amendments detailed in Annexes D and/or L.

Revision	Change Description	Date	Authority
1.0 (QSWB Feb 05)		04 Jul 05	DMTE
2.0		07 Dec 11	*O.S.B.* Cmdt CFNES
2.01		21 Mar 18	CO NTDC(P)
3.0		21 August 22	CO HMCS VENTURE
3.1	Adopt new RCN QS & TP format	16 June 23	CO HMCS VENTURE

NOTE:

Personnel programming courses must ensure that their hard copy of this document is current and identical to the electronic copy on the QSP SharePoint Site. See above for current changes.

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FOREWORD

1. This QSP is issued on the authority of the Commander of the Royal Canadian Navy and managed by the Commander Naval Personnel and Training Group (CNPTG) on behalf of the Assistant Chief of Naval Staff Personnel and Training (A/CNS P&T).

2. This QSP is based upon the Canadian Forces Manual of Individual Training and Education Naval Combat Systems Engineering MOSID 00344 Integrated Occupation Specification, 4 November 2004 A-PD-055-002/PP-001.

3. This QSP publication is effective on receipt and supersedes QSP revision 3.0, dated 22 August 2023. This QSP is designed to assist in developing, conducting, evaluating, and validating the Individual Training and Education (IT&E) required for training.

4. The QS at Chapters 1 and 2 were developed by the Qualification Standard Writing Board (QSWB) convened 8 Feb 05 at CFB Halifax working in collaboration with subject matter experts drawn from participating organizations. The subsequent TP at Chapters 3 and 4 were developed by the Training Plan Writing Board (TPWB) convened 8 Feb 05 at CFB Halifax working in collaboration with subject matter experts drawn from participating organizations. All aspects of this document were prepared in accordance with the concept of training outlined in A-P9-050 Series, Canadian Forces Manual of Individual Training and Education. Full details of the WBs are contained within Annex D and L.

5. The criteria for major and minor amendments, and the approval process, are identified in NPTORD 4500-22.

6. In circumstances where the QS and/or TP cannot be adhered to, submission of a Training Waiver in accordance with NPTORD 4500-22 is required.

QUALIFICATION STANDARD

CHAPTER 1 – QUALIFICATION STANDARD

AIM

1. The aim of this QSP is to establish the requirements and training for the Naval Combat Systema Engineering officer (NCS Eng) to perform their duties as a ship's/submarine's Combat Systems Engineering Officer. Possession of this qualification will allow personnel to be employed as the head of a Ship's Combat Systems Engineering Department.

GRANTING OF NQUAL

2. A trainee will achieve the NCS Eng Head of Department Qualification (AISF) qualification when the standards listed in the Performance Objectives (POs) have been met.

CO-QUALIFICATION

3. In addition to the performance objectives specific to this training, the following additional qualifications are included in this training program to meet the requirements of the job. The MITE and NQuals associated with these qualifications shall be granted upon completion of the training program and shall be listed within the course report and will be delivered IAW the following:

a. AISF, NCS ENG Head of Department; and

b. 114808.

PREREQUISITES

4. To be selected for this course the loading authority must ensure that trainees meet the following prerequisites:

a. NCS Eng Basic Occupation Qualification; and

b. National Qualification Code AIRX

CHAPTER 2 – PERFORMANCE OBJECTIVES

PO 001 Manage Department Personnel

1. Performance: Manage Departmental Personnel

2. Conditions:

   a. Given:

      (1) CSE Dept in a ship or submarine; and

      (2) References.

   b. Denied:

      (1) Supervision.

   c. Environment:

      (1) Alongside, at sea, under all conditions day or night.

3. Standard: IAW Para 4, the NCS Eng shall manage CS personnel by:

   a. Advising on Operational Limitations due to Personnel Shortages to include evaluating the impact of personnel limitations on operational requirements by determining departmental requirements to meet the operational schedule and addressing personnel shortfalls.

   b. Managing Individual Training and Education by:

      (1) Reviewing career progression and requirements for all occupations in the department;

      (2) Reviewing and maintaining training documentation;

      (3) Conducting individual interviews/counselling;

      (4) Liaising with career management and training personnel to achieve required training; and

      (5) Monitoring and conducting training.

   c. Providing divisional support for military personnel by:

      (1) Maintaining divisional/feedback notes;

(2) Producing administrative staff work;

(3) Counselling of subordinates' personal issues that impact their job performance;

(4) Acting as Assisting/Attending Officer;

(5) Evaluating subordinates through the PACE process; and

(6) Acknowledging superior performers via honours and awards programs and accelerated promotions.

d. Managing operational training including command team weapons system certification and by ensuring refresher training and team training are current to meet operational requirements by:

(1) Scheduling exercises to meet CRRs;

(2) Developing ERT policy and procedure; and

(3) Liaising with Training Chief/PO to determine training requirements

4. Reference: As per Annex A.

5. Task Numbers: BT5, BT10, BT15, CT10, CT15, CT50, CT55, ET15.

6. Remarks: Nil.

PO 002 Liaise with Scientific Establishments and Industry

1. Performance: Liaise with Scientific Establishments and Industry

2. Conditions:

   a. Given:

      (1) CSE Dept in a ship or submarine;

      (2) References; and

      (3) Direction from Formation authorities.

   b. Denied:

      (1) Supervision.

   c. Environment:

      (1) Alongside, at sea, under all conditions day or night.

3. Standard: IAW Para 4, the NCS Eng shall effectively liaise with laboratories, industries and establishments to support naval engineering such as:

   a. Defence Research and Development Canada (DRDC)

   b. NETE

   c. QETE

   d. OEM/FSR

4. Reference: As per Annex A.

5. Task Numbers: AT30.

6. Remarks: Nil.

PO 003 Manage Departmental Financial Resources

1. Performance: Manage Departmental Financial Resources

2. Conditions:

   a. Given:

      (1) CSE Dept in a ship or submarine;

      (2) References;

      (3) Departmental Bedget; and

      (4) FAA Section 32, Section 34 (Expenditure Management) Qualification.

   b. Denied:

      (1) Supervision.

   c. Environment:

      (1) Alongside, at sea, under all conditions day or night.

3. Standard: IAW Para 4, the NCS Eng shall manage the departmental budget and ensure compliance with procurement and contracting policy to include:

   a. Completion of the Expenditure Management course

   b. Reviewing policy

   c. Reviewing budget allocation

   d. Establishing priorities

   e. Tracking expenditures

   f. Ensuring appropriate items are properly accounted for and documented

4. Reference: As per Annex A.

5. Task Numbers: MT40, MT50.

6. Remarks: Nil.

PO 004 Manage Shipboard Risk

1. Performance: Manage Shipboard Risk

2. Conditions:

   a. Given:

      (1) CSE Dept in a ship or submarine;

      (2) References; and

      (3) Unit RF Safety Officer Qualification.

   b. Denied:

      (1) Supervision.

   c. Environment:

      (1) Alongside, at sea, under all conditions day or night.

3. Standard: IAW Para 4, the NCS Eng shall manage shipboard risk by:

   a. Promoting environmental awareness onboard ship by being a member of the Ship's general safety committee

   b. Manage Interdepartmental safety procedures including ensuring compliance with general safety policies and procedures by being a member of the Ship's general safety committee

   c. Perform the duties of the Unit RF Safety Officer

   d. Perform the duties of the head of the HAZMAT Cleanup Team

   e. Ensure compliance with policies/ governing procurement, stowage, handling, disposal and training of hazardous materials

4. Reference: As per Annex A.

5. Task Numbers: AT40, FT40, FT45, FT35, FT5, FT50.

6. Remarks: Nil.

PO 005 Prepare Technical Documentation

1. Performance: Prepare Technical Documentation

2. Conditions:

   a. Given:

      (1) CSE Dept in a ship or submarine; and

      (2) References.

   b. Denied:

      (1) Supervision.

   c. Environment:

      (1) Alongside, at sea, under all conditions day or night.

3. Standard: IAW Para 4, the NCS Eng shall prepare and review technical documentation to include:

   a. Completing Routine Engineering Reports such as:

      (1) Combat Readiness Requirement (CRR) Report

      (2) Monthly Ammunition Expenditure Report

      (3) Monthly Maintenance Reports

4. Reference: As per Annex A.

5. Task Numbers: AT5, DT105, GT15.

6. Remarks: Nil.

PO 006 Participate in Interdepartmental Operational Planning

1. Performance: Participate in Interdepartmental Operational Planning

2. Conditions:

   a. Given:

      (1) CSE Dept in a ship or submarine; and

      (2) References.

   b. Denied:

      (1) Supervision.

   c. Environment:

      (1) Alongside, at sea, under all conditions day or night.

3. Standard: IAW Para 4, the NCS Eng shall:

   a. Participate in planning ship's operations, program and/or routine to determine optimum system configurations by liaising with other heads of department to determine departmental requirements for the program

   b. Liaise with the MSEO to assess the impact of marine systems on combat systems

4. Reference: As per Annex A.

5. Task Numbers: GT70, DT125, DT135.

6. Remarks: Nil.

PO 007 Manage Signature Management Program

1. Performance: Manage Signature Management Program

2. Conditions:

   a. Given:

      (1) CSE Dept in a ship or submarine;

      (2) References; and

      (3) Signature Management Organization.

   b. Denied:

      (1) Supervision.

   c. Environment:

      (1) Alongside, at sea, under all conditions day or night.

3. Standard: IAW para 4, the NCS Eng shall perform the duties of the Ship's Signature Management Officer.

4. Reference: CFCD 109, NAVORD 3395-1.

5. Task Numbers: IT20, DT55, DT120.

6. Remarks: Nil.

PO 008 Manage Shipboard Configuration

1. Performance: Manage Shipboard Configuration;

2. Conditions:

   a. Given:

      (1) CSE Dept in a ship or submarine;

      (2) References; and

      (3) Approved As-Fitted drawings.

   b. Denied:

      (1) Supervision.

   c. Environment:

      (1) Alongside, at sea, under all conditions day or night.

3. Standard: IAW para 4, the NCS Eng shall manage configuration changes to include:

   a. Implementing approved configuration changes

   b. Initiating Engineering Change Proposals

   c. Preventing unauthourized changes to configured items

4. Reference: As per Annex A.

5. Task Numbers: HT35, LT40, LT55.

6. Remarks: Nil.

PO 009 Manage Test and Trials Program

1. Performance: Manage Test and Trials Program

2. Conditions:

   a. Given:

      (1) CSE Dept in a ship or submarine;

      (2) References;

      (3) Trial Agendas; and

      (4) Operational Schedule.

   b. Denied:

      (1) Supervision.

   c. Environment:

      (1) Alongside, at sea, under all conditions day or night.

3. Standard: IAW para 4 and relevant safety and environmental directives, the NCS Eng shall manage the Ship's tests and trials programme to include:

   a. Developing a test and trials programme through liaison with applicable technical and operational authourities

   b. Reviewing and commenting on trial agendas

   c. Conducting tests and trials

   d. Analyzing trial reports and advising command of results

   e. Taking necessary action to rectify defects

4. Reference: As per Annex A.

5. Task Numbers: KT40, KT20, KT10, KT5.

6. Remarks: Nil.

PO 010 Manage Readiness

1. Performance: Manage Readiness

2. Conditions:

   a. Given:

      (1) CSE Dept in a ship or submarine; and

      (2) References.

   b. Denied:

      (1) Supervision.

   c. Environment:

      (1) Alongside, at sea, under all conditions day or night.

3. Standard: IAW para 4, the NCS Eng shall:

   a. Maximize combat systems availability within the stated notice for power to include:

      (1) Assessing technical readiness and advising command on mission readiness by:

         (a) Monitoring the status of combat systems;

         (b) Assessing pre-sail checks;

         (c) Conducting engineering rounds;

         (d) Reviewing periodic equipment diagnostics and performance checks;

         (e) Assessing Departmental personnel readiness;

         (f) Preparing an OPDEF Summary message;

         (g) Analyzing the effects of equipment degradation and assessing impact on other systems;

         (h) Verifying the currency of maintenance and operating documentation; and

         (i) Verifying spare parts inventory.

(2) Directing maintenance activities.

b. Maximize combat systems availability at sea by:

(1) Advising command on mission readiness by:

(a) Assessing technical readiness;

(b) Monitoring of combat systems;

(c) Conducting engineering rounds;

(d) Reviewing logs and state boards;

(e) Assessing Departmental personnel readiness;

(f) Preparing an OPDEF summary report; and

(g) Analyzing the effects of equipment degradation and assessing impact on other systems.

(2) Directing maintenance activities.

c. Advise on Ship's States of Preparedness

d. Advise on Shipboard Degrees of Readiness

4. Reference: As per Annex A.

5. Task Numbers: DT130B, DT130, DT10, DT15, DT35, DT50, DT5, DT70, DT65.

6. Remarks: Nil.

PO 011 Manage Defect Rectification

1. Performance: Manage Defect Rectification

2. Conditions:

   a. Given:

      (1) CSE Dept in a ship or submarine; and

      (2) Operational Schedule.

   b. Denied:

      (1) Supervision.

   c. Environment:

      (1) Alongside, at sea, under all conditions day or night.

3. Standard: IAW para 4, the NCS Eng shall direct and supervise repair action due to equipment failures by:

   a. Receiving defect reports from operators and maintainers

   b. Investigating cause, nature and extent of equipment defects and failures

   c. Assessing impact of equipment failures on operations and advising command on operational limitations under various system operating modes

   d. Determining requirements for equipment and repair team safety and isolating system faults

   e. Identifying requirements and methods of obtaining spares

   f. Planning emergency repairs by taking command priorities and directing repair teams as appropriate

   g. Drafting and managing OPDEFs

4. Reference: As per Annex A.

5. Task Numbers: DT40, DT95, IT10, DT115, GT75, DT20, DT45.

6. Remarks: Nil.

PO 012 Manage Battle Damage

1. Performance: Manage Battle Damage
2. Conditions:
    a. Given:
        (1) CSE Dept in a ship or submarine;
        (2) References;
        (3) ERT Pack and ERT; and
        (4) Electronic ERT Pack (HFX).
    b. Denied:
        (1) Supervision.
    c. Environment:
        (1) Alongside, at sea, under all conditions day or night.
3. Standard: IAW para 4, the NCS Eng shall direct the Ship's ERT to include:
    a. Determining the status of the ERT when closed up
    b. Assessing combat systems readiness
    c. Monitoring equipment state/damage
    d. Receiving defect reports
    e. Comparing command capability priorities with reported defects
    f. Directing ERT to respond to the prioritized plan
4. Reference: As per Annex A.
5. Task Numbers: DT75, ET20, ET10.
6. Remarks: Nil.

PO 013 Respond to non-routine operations

1. Performance: Respond to non-routine operations

2. Conditions:

   a. Given:

      (1) CSE Dept in a ship or submarine; and

      (2) References.

   b. Denied:

      (1) Supervision.

   c. Environment:

      (1) Alongside, at sea, under all conditions day or night.

3. Standard: IAW para 4, the NCS Eng shall respond to non-routine emergencies to include:

   a. Directing a HAZMAT clean-up team by:

      (1) Ensuring duty watch, individual and team training are current; and

      (2) Ensuring necessary equipment is stocked and/or maintained.

   b. Responding to ammunition emergencies

4. Reference: As per Annex A.

5. Task Numbers: ET30, ET5.

6. Remarks: Nil.

PO 014 Manage Docking

1. Performance: Manage Docking

2. Conditions:

   a. Given:

      (1) CSE Dept in a ship or submarine; and

      (2) References.

   b. Denied:

      (1) Supervision.

   c. Environment:

      (1) Alongside, at sea, under all conditions day or night.

3. Standard: IAW Para 4, the NCS Eng shall prepare the ship for docking/undocking by meeting the departmental requirements of the docking/undocking memorandum.

4. Reference: As per Annex A.

5. Task Numbers: IT15.

6. Remarks: Nil.

PO 015 Manage Ship's Tiered Readiness Program (TRP)

1. Performance: Manage Ship's Tiered Readiness Program (TRP)

2. Conditions:

   a. Given:

      (1) CSE Dept in a ship or submarine;

      (2) References;

      (3) Appropriate repair facility support; and

      (4) Appropriate Formation support.

   b. Denied:

      (1) Supervision.

   c. Environment:

      (1) Alongside, at sea, under all conditions day or night.

3. Standard: IAW para 4, the officer shall manage the departmental responsibilities during a TRP to include:

   a. Managing equipment readiness

   b. Managing personnel training

4. Reference: As per Annex A.

5. Task Numbers: TBD.

6. Remarks: Nil.

PO 016 Manage Formation Level Maintenance Activities

1. Performance: Manage Formation Level Maintenance Activities)

2. Conditions:

   a. Given:

      (1) CSE Dept in a ship or submarine;

      (2) References;

      (3) Current naval maintenance management information systems;

      (4) Appropriate repair facility support; and

      (5) Appropriate Formation support.

   b. Denied:

      (1) Supervision.

   c. Environment:

      (1) Alongside, at sea, under all conditions day or night.

3. Standard: IAW para 4, the NCS Eng shall manage formation level maintenance activities on ship by:

   a. Preparing and staffing first line O&M documentation to include:

      (1) Identifying deficiencies in maintenance schedules through the UCR process;

      (2) Developing new maintenance procedures and techniques;

      (3) Evaluating new maintenance procedures and techniques; and

      (4) Reviewing and approving MAFs and associated stores demands.

   b. Using naval maintenance management information systems to include:

      (1) Identifying deficiencies in maintenance schedules through the UCR process;

      (2) Developing new maintenance procedures and techniques;

      (3) Evaluating new maintenance procedures and techniques; and

(4) Reviewing DRMIS notifications and associated stores demands.

c. Liaising with Formation staff with respect to priority to include:

(1) Identifying/referring maintenance tasks beyond ship's staff capability; and

(2) Prioritizing maintenance action to include:

(a) Reviewing corrective maintenance;

(b) Reviewing current preventive maintenance; and

(c) Reviewing outstanding preventive maintenance.

d. Monitor contact work to include:

(1) Contract progress against scheduled milestones;

(2) Technical compliance with contracts; and

(3) Reviewing contract deliverables for compliance.

e. Manage Test Equipment Maintenance Management Information System (TEMMIS) to include

(1) Reviewing TEMMIS reports; and

(2) Identifying TEMMIS requirements.

4. Reference: As per Annex A.

5. Task Numbers: HT40, LT80, HT30, GT100, LT75, GT40, DT110, HT20, GT45, DT115, HT25.

6. Remarks: Nil.

PO 017 Manage a Short Work Period (SWP)

1. Performance: Manage a Short Work Period (SWP)

2. Conditions:

   a. Given:

      (1) CSE Dept in a ship or submarine;

      (2) References;

      (3) Appropriate repair facility support; and

      (4) Appropriate Formation support.

   b. Denied:

      (1) Supervision.

   c. Environment:

      (1) Alongside, at sea, under all conditions day or night.

3. Standard: IAW para 4, the NCS Eng shall manage a SWP to include:

   a. Preparing for the short work period

   b. Managing the short work period

   c. Completing the short work period

4. Reference: As per Annex A.

5. Task Numbers: TBD.

6. Remarks: Nil.

PO 018 Manage ADM (Mat) Level Maintenance Activities

1. Performance: Manage ADM (Mat) Level Maintenance Activities

2. Conditions:

   a. Given:

      (1) CSE Dept in a ship or submarine;

      (2) References;

      (3) Appropriate ADM (Mat) support; and

      (4) Appropriate Formation support.

   b. Denied:

      (1) Supervision.

   c. Environment:

      (1) Alongside, at sea, under all conditions day or night.

3. Standard: IAW para 4, the NCS Eng shall manage ADM(Mat) level maintenance activities on ship to include:

   a. Planning a refit/docking work period to include pre-refit/DWP trials and surveys

   b. Preparing and staffing 3rd line O&M documentation

   c. Monitoring contract work to include:

      (1) Contract progress against scheduled milestones;

      (2) Observation reports for defects/anomalies during refits/DWPs/Ship; construction

      (3) Technical compliance with contracts; and

      (4) Reviewing contract deliverables for compliance.

   d. Providing support to Quality Assurance Representatives to include:

      (1) Monitoring contract progress against scheduled milestones;

(2) Drafting observation reports for defects/anomalies during refits/DWPs/Ship construction;

(3) Ensuring technical compliance with contracts; and

(4) Reviewing contract deliverables for compliance.

4. Reference: As per Annex A.

5. Task Numbers: JT40, JT5, MT20, MT30, JT30, NT120, MT25, NT150, MT35, NT155.

6. Remarks: Nil.

PO 019 Manage Shipboard Ammunition and Explosive Safety

1. Performance: Manage Shipboard Ammunition and Explosive Safety

2. Conditions:

   a. Given:

      (1) CSE Dept in a ship or submarine; and

      (2) References.

   b. Denied:

      (1) Supervision.

   c. Environment:

      (1) Alongside, at sea, under all conditions day or night.

3. Standard: IAW para 4, the NCS Eng shall:

   a. Describe ammunition organizations, both ashore and onboard

   b. Manage the storage and handling of ammunition

   c. Perform the duties of the shipboard Ammunition Officer

   d. Manage the shipboard Explosive Safety Program

4. Reference: As per Annex A.

5. Task Numbers: FT15.

6. Remarks: Nil.

PO 020 Manage Departmental Systems

1. Performance: Manage Departmental Systems

2. Conditions:

   a. Given:

      (1) CSE Dept in a ship or submarine; and

      (2) References.

   b. Denied:

      (1) Supervision.

   c. Environment:

      (1) Alongside, at sea, under all conditions day or night.

3. Standard: IAW para 4, the NCS Eng shall perform the following duties:

   a. Manage Above Water Sensor systems to include:

      (1) Monitoring start-up, running and shutdown of systems;

      (2) Reviewing system logs and operating records;

      (3) Monitoring diagnostic tests;

      (4) Monitoring equipment performance checks; and

      (5) Advising command on system status.

   b. Manage Communications systems to include:

      (1) Monitoring start-up, running and shutdown of systems;

      (2) Reviewing system logs and operating records;

      (3) Monitoring diagnostic tests;

      (4) Monitoring equipment performance checks; and

      (5) Advising command on system status.

   c. Manage Command and Control systems to include:

      (1) Monitoring start-up, running and shutdown of systems;

(2) Reviewing system logs and operating records;

(3) Monitoring diagnostic tests;

(4) Monitoring equipment performance checks; and

(5) Advising command on system status.

d. Manage Underwater Sensor systems to include:

(1) Monitoring start-up, running and shutdown of systems;

(2) Reviewing system logs and operating records;

(3) Monitoring diagnostic tests;

(4) Monitoring equipment performance checks; and

(5) Advising command on system status.

e. Manage Electronic Warfare systems to include:

(1) Monitoring start-up, running and shutdown of systems;

(2) Reviewing system logs and operating records;

(3) Monitoring diagnostic tests;

(4) Monitoring equipment performance checks; and

(5) Advising command on system status.

f. Manage Data Processing systems to include:

(1) Monitoring start-up, running and shutdown of systems;

(2) Reviewing system logs and operating records;

(3) Monitoring diagnostic tests;

(4) Monitoring equipment performance checks; and

(5) Advising command on system status.

g. Manage Navigation systems to include :

(1) Monitoring start-up, running and shutdown of systems;

(2) Reviewing system logs and operating records;

(3) Monitoring diagnostic tests;

(4) Monitoring equipment performance checks; and

(5) Advising command on system status.

h. Manage Weapon systems to include :

(1) Monitoring start-up, running and shutdown of systems;

(2) Reviewing system logs and operating records;

(3) Monitoring diagnostic tests;

(4) Monitoring equipment performance checks;

(5) Advising command on system status;

(6) Monitoring of pre-firing/post-firing checks; and

(7) Performing the duties of ship's Ammunition Officer.

4. Reference: As per Annex A.

5. Task Numbers: OT40, OT155, OT140, OT145, OT150, GT55, JT40, GT60, JT35, GT50, GT65.

6. Remarks: Nil.

TRAINING PLAN

CHAPTER 3 – TRAINING MANAGEMENT DETAILS

TRAINING STRATEGY

1. The following training strategies were selected for this course:

   a. On the job training;

TRAINING DURATION

2. Based on existing resources and approved instructional design requirements, the approved Course Design authorizes 365 training days.

REQUIRED COURSE STAFFING

3. There are no required staffing as this is an on the job training course conducted on an HMC Ship.

TRAINING LIMITATION

4. A training limitation may be due to constraints of resources, personnel or equipment. A training limitation is the difference between what a qualified individual must be able to do and what training can prepare them to do. The following limitations are identified below:

   a. Nil.

COURSE CAPACITY

5. The capacity per session is based on the guidelines in CFITES Vol 5 – Instructor to Learner Ratios:

   a. Optimum – is 1 trainee as this course does not have any class lectures.

   b. Maximum Load – is 1 trainee as this course does not have any class lectures

   c. Minimum – is 1 trainee as this course does not have any class lectures

DESIGNATED SCHEDULING AND COURSE LOADING

6. Training will be scheduled utilizing the HRMS Training (MITE) module to meet the training requirements (IAW NPTORD 4540-2).

7. The authority to load members onto this course will be IAW NPTORD 4540-43. NPTG HQ is to be included as an info addressee on all loading messages. Course loading will be accomplished by:

a. The Career Manager

QUALIFICATION AND ASSESSMENT

8. This is a pass/fail course. Students must successfully complete all POs to achieve the qualification.

9. Students will be monitored and assessed both on performance and conduct throughout the course. Counselling, progress monitoring, and maintenance of student files will be in accordance with NPTORD 4500-55. Although not mandated specifically in the POs, the general demands of service members and the leadership demands of junior and senior leaders form an integral part of individual training. As such, students are presumed to possess the leader and/or follower attributes commensurate with their rank.

UNSATISFACTORY PROGRESS

10. Any student who fails to successfully complete all POs in the allocated number of attempts will be referred to the Commandant (Cmdt) or Commanding Officer (CO) of the training establishment (TE) to determine the requirement for a TRB IAW NPTORD 4500-54.

REMOVAL FROM TRAINING

11. A TE CO/Cmdt may direct that a student be removed from training for reasons of deficiency in performance, conduct (discipline, attitude, teamwork, safety, leadership), missed training, medical, and other non-performance/conduct related issues. In these cases, Cease Training Procedures are to be strictly adhered to IAW NPTORD 4500-29 or 4500-54.

RECERTIFICATION REQUIREMENTS

12. There are no additional requirements to maintain currency in this qualification.

LANGUAGE OF INSTRUCTION

13. The language of instruction shall be in accordance with the Delivery of Training and Education in Both Official Languages (DAOD 5039-6).

PRIOR LEARNING ASSESSMENT AND RECOGNITION (PLAR)

14. PLARs shall be conducted in accordance with DAOD 5031-1 Canadian Forces Military Equivalencies Program, through the RCN PLAR submissions coordinator.

POST COURSE REVIEWS (PCR)

15. PCRs are to be conducted on all steady state courses IAW NPTORD 4500-23.

COURSE REPORTING

16. Formal Course: A Canadian Forces Course Report, Form CF 377 and a CF289 Certificate of Military Achievement, shall be prepared and distributed in accordance with DAOD 5031-9. The Course Report shall be descriptive, identifying the student's strengths and weaknesses with specific comments on any PO for which the student demonstrated outstanding performance or had difficulty achieving the standard, any conduct relevant to course performance, and any conditions or situations that affected course performance.

CHAPTER 4 LESSON SPECIFICATIONS

PO 001 – MANAGE DEPARTMENTAL PERSONNEL

EO 001.01 ADVISE ON OPERATIONAL LIMITATIONS DUE TO PERSONNEL SHORTAGES

1. Performance: Advise on Operational Limitations due to Personnel Shortages

2. Conditions:

   a. Given:

      (1) CSE Dept in a ship or submarine; and

      (2) References.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Alongside, at sea, under all conditions day or night.

3. Standard: IAW Refs, the NCS Eng shall evaluate the impact of personnel limitations on operational requirements to include:

   a. Determining departmental requirements to meet the operational schedule by:

      (1) Reviewing AER;

      (2) Review ship's program;

      (3) Liaise with CSE Chief regarding Watch and Station Bill; and

      (4) Identify and determine SUBSAFE Personnel Pillar implications and mitigations in response to Personnel Shortages (if applicable).

   b. Addressing personnel shortfalls by

      (1) Briefing the XO on personnel readiness; and

      (2) Draft a personnel shortage message.

4. Teaching Points: N/A

5. Total EO Time: Throughout 12 months of training time.

6. Training and Learning Aids: Nil.

7. Remarks: Nil.

PO 001 – MANAGE DEPARTMENTAL PERSONNEL

EO 001.02 MANAGE INDIVIDUAL TRAINING AND EDUCATION

1. Performance: Manage Individual Training and Education

2. Conditions:

   a. Given:

      (1) CSE Dept in a ship or submarine; and

      (2) References.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Alongside, at sea, under all conditions day or night.

3. Standard: IAW refs, the NCS Eng shall manage individual training and education by:

   a. Reviewing career progression and requirements for all occupations in the department

   b. Reviewing and maintaining training documentation

   c. Conducting individual interviews/counselling

   d. Liaising with career management and training personnel to achieve required training

   e. Monitoring and conducting training to include:

      (1) Prioritizing training requirements;

      (2) Organizing and assessing training opportunities; and

      (3) Participating in pre-boards and qualification boards.

4. Teaching Points: N/A

5. Total EO Time: Throughout 12 months of training time.

6. Training and Learning Aids: Nil.

7. Remarks: Nil.

PO 001 – MANAGE DEPARTMENTAL PERSONNEL

EO 001.03 PROVIDE DIVISIONAL SUPPORT FOR MILITARY PERSONNEL

1. Performance: Provide Divisional support for military personnel

2. Conditions:

   a. Given:

      (1) CSE Dept in a ship or submarine; and

      (2) References.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Alongside, at sea, under all conditions day or night.

3. Standard: IAW refs, the NCS Eng shall perform the duties of a Divisional Officer to include:

   a. Maintaining divisional notes

   b. Producing administrative staff work

   c. Counselling of subordinates' personal issues that impact on job performance, to include:

      (1) Administrative measures;

      (2) Drugs, alcohol and harassment;

      (3) Financial issues;

      (4) Sexual harassment and/or misconduct;

      (5) Advise on alternate career opportunities including commissioning plans, occupational transfers and release; and

      (6) Personal relationships.

4. Teaching Points: N/A

5. Total EO Time: Throughout 12 months of training time.

6. Training and Learning Aids: Nil.

7. Remarks: Nil.

PO 001 – MANAGE DEPARTMENTAL PERSONNEL

EO 001.04 MANAGE OPERATIONAL TRAINING

1. Performance: Manage operational training

2. Conditions:

   a. Given:

      (1) CSE Dept in a ship or submarine; and

      (2) References.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Alongside, at sea, under all conditions day or night.

3. Standard: IAW refs, the NCS Eng shall manage operational training to include:

   a. Participating in :

      (1) Command team weapons system certification;

      (2) Submarine Weapons System Team Training (if applicable); and

      (3) Submarine Weapons Handling and Discharge Team Training (if applicable).

   b. Ensuring refresher training and team training are current to meet operational requirements by:

      (1) Scheduling exercises to meet CRRs;

      (2) Developing ERT policy and procedure through liaison with Sea Training staff and training establishments;

      (3) Liaising with Training Chief/PO to determine training requirements; and

      (4) Working with Operations Officer to support/meet Weapons Certification process and training.

4. Teaching Points: N/A

5. Total EO Time: Throughout 12 months of training time.

6. Training and Learning Aids: Nil.

7. Remarks: Nil.

PO 002 – LIAISE WITH SCIENTIFIC ESTABLISHMENTS AND INDUSTRY

EO 002.01 LIAISE WITH SCIENTIFIC ESTABLISHMENTS AND INDUSTRY

1. Performance: Liaise with Scientific establishments and Industry

2. Conditions:

   a. Given:

      (1) CSE Dept in a ship or submarine;

      (2) References; and

      (3) Direction from Formation authorities.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Alongside, at sea, under all conditions day or night.

3. Standard: IAW refs, the NCS Eng shall demonstrate comprehensive knowledge of the roles and capabilities of laboratories, industries and establishments, and the procedures to request their assistance, in order to support naval engineering. These units can include:

   a. Defence Research and Development Canada (DRDC)

   b. NETE

   c. QETE

   d. OEM/FSR

4. Teaching Points: N/A

5. Total EO Time: Throughout 12 months of training time.

6. Training and Learning Aids: Nil.

7. Remarks: Nil.

PO 003 – MANAGE DEPARTMENTAL FINANCIAL RESOURSES

EO 003.01 MANAGE DEPARTMENTAL BUDGET AND ENSURE COMPLIANCE WITH PROCUREMENT AND CONTRACTING POLICY

1. Performance: Manage the departmental budget and ensure compliance with procurement and contracting policy

2. Conditions:

   a. Given:

      (1) CSE Dept in a ship or submarine;

      (2) References; and

      (3) Departmental Budget.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Alongside, at sea, under all conditions day or night.

3. Standard: IAW refs, the NCS Eng manage the departmental budget and ensure compliance with procurement and contracting policy to include::

   a. Completion of the Expenditure Management course

   b. Reviewing policy to include unit level direction on budget management

   c. Reviewing budget allocation

   d. Establishing priorities

   e. Tracking expenditures

   f. Ensuring appropriate items are properly accounted for and documented

   g. Letting a contract in a foreign port

4. Teaching Points: N/A

5. Total EO Time: Throughout 12 months of training time.

6. Training and Learning Aids: Nil

7. Remarks: Nil

PO 004 – MANAGE SHIPBOARD RISK

EO 004.01 PROMOTE ENVIRONMENTAL AWARENESS ONBOARD SHIP

1. Performance: Promote environmental awareness onboard ship

2. Conditions:

   a. Given:

      (1) CSE Dept in a ship or submarine; and

      (2) References.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Alongside, at sea, under all conditions day or night.

3. Standard: IAW refs, the NCS Eng shall promote environmental awareness onboard ship by being a member of the Ship's Environmental Committee.

4. Teaching Points: N/A

5. Total EO Time: Throughout 12 months of training time.

6. Training and Learning Aids: Nil.

7. Remarks: Nil.

PO 004 – MANAGE SHIPBOARD RISK

EO 004.02 MANAGE INTERDEPARTMENTAL SAFETY PROCEDURES

1. Performance: Manage interdepartmental safety procedures

2. Conditions:

   a. Given:

      (1) CSE Dept in a ship or submarine; and

      (2) References.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Alongside, at sea, under all conditions day or night.

3. Standard: IAW refs, the NCS Eng shall manage interdepartmental safety procedures to include:

   a. Ensuring compliance with general safety policies and procedures by:

      (1) Assisting in the completion of a SPDET;

      (2) Completing monthly safety rounds;

      (3) Completing CF98s and DND663s; and

      (4) Completing LOTO Audits.

   b. Participating in the Ship's general safety committee

4. Teaching Points: N/A

5. Total EO Time: Throughout 12 months of training time.

6. Training and Learning Aids: Nil.

7. Remarks: Nil.

PO 004 – MANAGE SHIPBOARD RISK

EO 004.03 PERFORMS AS UNIT RF SAFETY OFFICER

1. Performance: Perform as Unit RF Safety Officer

2. Conditions:

    a. Given:

        (1) CSE Dept in a ship or submarine; and

        (2) References.

    b. Denied:

        (1) Nil.

    c. Environment:

        (1) Alongside, at sea, under all conditions day or night.

3. Standard: IAW refs, the NCS Eng shall perform the duties of the Unit RF Safety Officer to include:

    a. Completion of the Unit RF Safety Officer course

    b. Conducting shipboard RF Safety training

    c. Attending formation level meetings and briefings concerning RF Safety

    d. Performing the duties of Ship's Assistant Unit RF Safety Officer

    e. Completion of periodic RF Safety Audits as required by SEMs

    f. Initiating investigations into RF Safety incidents to include:

        (1) Drafting RF Safety incident messages; and

        (2) Requirements for further investigation.

    g. Reviewing results of investigation and take appropriate actions

    h. Reviewing RF Safety incident messages

4. Teaching Points: N/A

5. Total EO Time: Throughout 12 months of training time.

6. Training and Learning Aids: Nil.

7. Remarks: Nil.

PO 004 – MANAGE SHIPBOARD RISK

EO 004.04 PERFORMS THE DUTIES OF THE HEAD OF THE HAZMAT CLEANUP TEAM

1. Performance: Perform the duties of the head of the HAZMAT Cleanup Team

2. Conditions:

   a. Given:

      (1) CSE Dept in a ship or submarine; and

      (2) References.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Alongside, at sea, under all conditions day or night.

3. Standard: IAW refs, the NCS Eng shall perform the duties of the head of the HAZMAT Cleanup Team to include:

   a. Ensuring duty watch, individual and team training are current

   b. Ensuring necessary equipment is stocked and/or maintained

   c. Carrying out HAZMAT spill response exercises

   d. Describe the HAZMAT Cleanup organization in a submarine (if applicable)

   e. Act as CSEO in HQ1 during a submarine HAZMAT exercise (if applicable)

4. Teaching Points: N/A

5. Total EO Time: Throughout 12 months of training time.

6. Training and Learning Aids: Nil.

7. Remarks: This enabling objective is directly related to EO 013.01.

PO 004 – MANAGE SHIPBOARD RISK

EO 004.05 ENSURE COMPLIANCE WITH POLICIES/PROCESURES GOVERNING HANDLING, STOWAGE AND DISPOSAL OF HAZARDOUS MATERIALS

1. Performance: Ensure compliance with policies/procedures governing handling, stowage and disposal of hazardous materials

2. Conditions:

   a. Given:

      (1) CSE Dept in a ship or submarine; and

      (2) References.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Alongside, at sea, under all conditions day or night.

3. Standard: IAW refs, the NCS Eng shall ensure compliance with policies/procedures governing procurement, stowage, handling, disposal and product-specific training for hazardous materials.

4. Teaching Points: N/A

5. Total EO Time: Throughout 12 months of training time.

6. Training and Learning Aids: Nil.

7. Remarks: Nil.

PO 005 – PREPARE TECHNICAL DOCUMENTATION

EO 005.01 COMPLETE ROUTINE ENGINEERING REPORTS

1. Performance: Complete Routine Engineering Reports

2. Conditions:

   a. Given:

      (1) CSE Dept in a ship or submarine; and

      (2) References.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Alongside, at sea, under all conditions day or night.

3. Standard: IAW refs, the NCS Eng shall complete routine engineering reports such as:

   a. Signature Management Reports

   b. Combat Readiness Requirement (CRR) Report

   c. Monthly Maintenance Reports

   d. Monthly Ammunition Expenditure Report

4. Teaching Points: N/A

5. Total EO Time: Throughout 12 months of training time.

6. Training and Learning Aids: Nil.

7. Remarks: Nil.

PO 005 – PREPARE TECHNICAL DOCUMENTATION

EO 005.02 REVIEW AND MAINTAIN TECHNICAL DIRECTIVES, ORDERS AND RECORDS

1. Performance: Review and maintain technical directives, orders and records

2. Conditions:

   a. Given:

      (1) CSE Dept in a ship or submarine; and

      (2) References.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Alongside, at sea, under all conditions day or night.

3. Standard: IAW refs, the NCS Eng shall review and maintain technical directives, orders and records such as:

   a. Ship's SOPs

   b. COTMs

   c. CSEOTIs

   d. CSEOTMs

   e. Captain's Ship Book

   f. VDS log

   g. CANTASS towing log

   h. Gyro log

   i. Supersession

   j. SUBSAFE Directives and SUBSAFE Notifications

   k. Lifting Appliance Certificates

   l. Quality Assurance Live File

m. Submarine First Level documents

n. Submarine Licenses and License Exceptions

o. Submarine Registries (Pressure Vessel, Flex Hose, First Level Fasteners)

4. Teaching Points: N/A

5. Total EO Time: Throughout 12 months of training time.

6. Training and Learning Aids: Nil.

7. Remarks: Nil.

PO 005 – PREPARE TECHNICAL DOCUMENTATION

EO 005.03 PREPARE MARITIME TECHNICAL POLICY

1. Performance: Prepare maritime technical policy

2. Conditions:

   a. Given:

      (1) CSE Dept in a ship or submarine; and

      (2) References.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Alongside, at sea, under all conditions day or night.

3. Standard: IAW refs, the NCS Eng shall prepare maritime technical policy by reviewing, drafting and commenting upon documents such as:

   a. NAVORDS/MARLANTORDS/MARPACORDS

   b. Fleet SOPs

   c. Technical Bulletins

   d. CFTOs

   e. NEM

   f. NaMMS

   g. Submarine Technical Instructions (SMTI)

   h. Victoria Class Submarine Technical Instructions (VSTI)

   i. Submarine Weapon Operating Procedures (SUBWOPs)

   j. Submarine Books of Reference (BRF)

4. Teaching Points: N/A

5. Total EO Time: Throughout 12 months of training time.

6. Training and Learning Aids: Nil.

7. Remarks: Nil.

PO 006 – PARTICIPATE IN INTERDEPARTMENTAL OPERATIONAL PLANNING

EO 006.01 PARTICIPATE IN PLANNING SHIP'S OPERATIONS, PROGRAM AND/OR ROUTINE

1. Performance: Participate in planning ship's operations, program and/or routine

2. Conditions:

   a. Given:

      (1) CSE Dept in a ship or submarine; and

      (2) References.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Alongside, at sea, under all conditions day or night.

3. Standard: IAW refs, the NCS Eng shall participate in planning ship's operations, program and/or routine to include:

   a. Liaising with other heads of department to provide input for the long and short term programs (ie. SAR, SUBSAR, long term deployment, force protection, TRP, Missile EX, ASWEX, CASEX, TORPEX) to include:

      (1) Test and Trials requirements;

      (2) Training requirements; and

      (3) Maintenance requirements.

   b. Liaising with other heads of department to provide engineering support by reacting to the ship's long and short term programs (ie. SAR, SUBSAR, long term deployment, force protection, TRP, Missile EX, ASWEX, CASEX, TORPEX) with respect to:

      (1) Mission requirements;

      (2) Maintenance requirements;

      (3) Training requirements; and

(4) Optimizing system configuration.

c. Participating in:

(1) HODs and CHODs;

(2) XO's Morning Prayers;

(3) In Port Training (IPT); and

(4) CO's Evening Brief.

4. Teaching Points: N/A

5. Total EO Time: Throughout 12 months of training.

6. Training and Learning Aids: Nil.

7. Remarks: Nil.

PO 006 – PARTICIPATE IN INTERDEPARTMENTAL OPERATIONAL PLANNING

EO 006.02 ASSESS IMPACT OF MARINE SYSTEMS ON COMBAT SYSTEMS

1. Performance: Liaise with MSEO to assess impact of marine systems on combat systems

2. Conditions:

   a. Given:

      (1) CSE Dept in a ship or submarine; and

      (2) References.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Alongside, at sea, under all conditions day or night.

3. Standard: IAW refs, the NCS Eng shall liaise with MSEO to assess the impact on combat systems of the loss or limitation of marine systems, to include but not be limited to:

   a. Power Generation and Distribution;

   b. HVAC;

   c. LP/HP Air;

   d. Chilled Water;

   e. Firemain/Sea Water Services;

   f. Fitted Firefighting equipment; and

   g. Hydraulics.

4. Teaching Points: N/A

5. Total EO Time: Throughout 12 months of training time.

6. Training and Learning Aids: Nil.

7. Remarks: Nil.

PO 007 – MANAGE SIGNATURE MANAGEMENT PROGRAM

EO 007.01 PERFORM DUTIES OF SHIP'S SIGNATURE MANAGEMENT OFFICER

1. Performance: Perform duties of Ship's Signature Management Officer

2. Conditions:

   a. Given:

      (1) CSE Dept in a ship or submarine;

      (2) References; and

      (3) Signature Management Organization.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Alongside, at sea, under all conditions day or night.

3. Standard: IAW refs, the NCS Eng shall perform duties of the Ship's Signature Management Officer to include:

   a. Applying signature management policies and procedures by:

      (1) Conducting signature management rounds and ranging;

      (2) Reviewing results from rounds and ranging;

      (3) Briefing the CO on signature management rounds and ranging; and

      (4) Reporting and rectifying defects through OPDEFs and corrective maintenance.

   b. Maintaining signature management reports

   c. Providing signature management reports bi-annually to the formation signature management officer

4. Teaching Points: N/A

5. Total EO Time: Throughout 12 months of training time.

6. Training and Learning Aids: Nil.

7. Remarks: Nil.

PO 008 – MANAGE SHIPBOARD CONFIGURATION

EO 008.01 CONTROL CONFIGURATION MANAGEMENT ONBOARD SHIP

1. Performance: Control configuration management onboard ship

2. Conditions:

   a. Given:

      (1) CSE Dept in a ship or submarine;

      (2) References; and

      (3) Approved As-Fitted drawings.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Alongside, at sea, under all conditions day or night.

3. Standard: IAW refs, the NCS Eng shall control configuration management onboard ship to include managing configuration changes by:

   a. Implementing approved configuration changes by liaising with formation

   b. Initiating Engineering Change Proposals including supporting documentation such as UCRs and SOCDs

   c. Preventing unauthorized changes to configured items

4. Teaching Points: N/A

5. Total EO Time: Throughout 12 months of training time.

6. Training and Learning Aids: Nil.

7. Remarks: Nil.

PO 009 – MANAGE TEST AND TRIALS PROGRAMME

EO 009.01 MANAGE SHIP'S TESTS AND TRIALS PROGRAMME

1. Performance: Manage Ship's tests and trials programme

2. Conditions:

   a. Given:

      (1) CSE Dept in a ship or submarine;

      (2) References;

      (3) Trial Agendas; and

      (4) Operational Schedule.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Alongside, at sea, under all conditions day or night.

3. Standard: IAW refs, the NCS Eng shall manage Ship's tests and trials programme to include:

   a. Developing a test and trials programme through liaison with applicable technical and operational authorities

   b. Reviewing and commenting on trial agendas

   c. Conducting tests and trials such as:

      (1) SPFs;

      (2) Surface Trials;

      (3) Static and Dynamic Sound Ranging;

      (4) CSAV; and

      (5) Harbour acceptance trials.

   d. Plan CSED contribution to Submarine trials programs to include:

      (1) Camber Dive;

(2) Surface Trials;

(3) Dived Trials;

(4) Initial Deep Dive; and

(5) Sonar, Weapons, and Ranging trials.

e. Analyzing trial reports and advising command of results

f. Taking necessary action to rectify defects

4. Teaching Points: N/A

5. Total EO Time: Throughout 12 months of training time.

6. Training and Learning Aids: Nil.

7. Remarks: Nil.

PO 010 – MANAGE READINESS

EO 010.01 MAXIMIZE COMBAT SYSTEMS AVAILABILITY WITHIN THE STATED NFP

1. Performance: Maximize combat systems availability within the stated notice for power

2. Conditions:

   a. Given:

      (1) CSE Dept in a ship or submarine; and

      (2) References.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Alongside, at sea, under all conditions day or night.

3. Standard: IAW refs, the NCS Eng shall maximize combat systems availability within the stated notice for power to include:

   a. Assessing technical readiness and advising command on mission readiness by:

      (1) Monitoring the status of combat systems;

      (2) Assessing pre-sail checks;

      (3) Conducting engineering rounds to include:

         (a) Ensuring departmental spaces are secured for sea;

         (b) Ensuring departmental spaces are free of hazards; and

         (c) Ensuring departmental spaces contain proper firefighting/damage control/safety equipment.

      (4) Reviewing periodic equipment diagnostics and performance checks;

      (5) Assessing Departmental personnel readiness;

      (6) Preparing an OPDEF summary report;

(7) Analyzing the effects of equipment degradation and assessing impact on other systems;

(8) Verifying the currency of maintenance and operating documentation; and

(9) Verifying departmental stores inventory.

b. Directing maintenance activities

4. Teaching Points: N/A

5. Total EO Time: Throughout 12 months of training time.

6. Training and Learning Aids: Nil.

7. Remarks: Nil.

PO 010 – MANAGE READINESS

EO 010.02 MAXIMIZE COMBAT SYSTEMS AVAILABILITY AT SEA

1. Performance: Maximize combat systems availability at sea

2. Conditions:

   a. Given:

      (1) CSE Dept in a ship or submarine; and

      (2) References.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Alongside, at sea, under all conditions day or night.

3. Standard: IAW refs, the NCS Eng shall maximize combat systems availability at sea to include:

   a. Assessing technical readiness and advising command on mission readiness by:

      (1) Monitoring the status of combat systems;

      (2) Conducting engineering rounds:

         (a) Ensuring departmental spaces are secured for sea;

         (b) Ensuring departmental spaces are free of hazards; and

         (c) Ensuring departmental spaces contain proper fire fighting/damage control/safety equipment.

      (3) Reviewing periodic equipment diagnostics and performance checks;

      (4) Reviewing logs and state boards;

      (5) Assessing Departmental personnel readiness;

      (6) Preparing an OPDEF summary report; and

(7) Analyzing the effects of equipment degradation and assessing impact on other systems.

4. Teaching Points: N/A

5. Total EO Time: Throughout 12 months of training time.

6. Training and Learning Aids: Nil.

7. Remarks: Nil.

PO 010 – MANAGE READINESS

EO 010.03 ADVISE ON SHIP'S STATES OF PREPAREDNESS

1. Performance: Advise on Ship's States of Preparednes

2. Conditions:

   a. Given:

      (1) CSE Dept in a ship or submarine; and

      (2) References.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Alongside, at sea, under all conditions day or night.

3. Standard: IAW refs, the NCS Eng shall advise the on watch ORO (or submarine dived OOW) on Ship's States of Preparedness to include:

   a. Demonstrating a complete understanding of the departmental personnel/technical requirements of the States of Preparedness to include:

      (1) Standing To;

      (2) Relaxed;

      (3) Stood Down; and

      (4) Secured.

   b. Ensuring the effective employment of CSE personnel and the preservation of equipment

   c. Submarine Weapons Handling and Discharge System Readiness States IAW SUBWOPs (if required)

   d. Submarine Submerged Signal Ejector Readiness States IAW SUBWOPs (if required)

4. Teaching Points: N/A

5. Total EO Time: Throughout 12 months of training time.

6. Training and Learning Aids: Nil.

7. Remarks: Nil.

PO 010 – MANAGE READINESS

EO 010.04 ADVISE ON SHIPBOARD DEGREES OF READINESS

1. Performance: Advise on Shipboard Degrees of Readiness

2. Conditions:

   a. Given:

      (1) CSE Dept in a ship or submarine; and

      (2) References.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Alongside, at sea, under all conditions day or night.

3. Standard: IAW refs, the NCS Eng shall:

   a. Manage the employment of departmental personnel to meet all Shipboard Degrees of Readiness using the following resources:

      (1) Watch and Station Bill;

      (2) CSE Chief and Cox'n to resolve conflicts in the WSB; and

      (3) Requirements of special parties.

   b. Advise command on the capabilities and limitations on departmental personnel to meet the required degrees of readiness

   c. Demonstrate an understanding of the Combat Information Organization in submarines (if applicable)

4. Teaching Points: N/A

5. Total EO Time: Throughout 12 months of training time.

6. Training and Learning Aids: Nil.

7. Remarks: Nil.

PO 011 – MANAGE DEFECT RECTIFICATION

EO 011.01 DIRECT AND SUPERVISE REPAIR ACTION DUE TO EQUIPMENT FAILURES

1. Performance: Direct and supervise repair action due to equipment failures

2. Conditions:

   a. Given:

      (1) CSE Dept in a ship or submarine; and

      (2) References.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Alongside, at sea, under all conditions day or night.

3. Standard: IAW ref, the NCS Eng shall direct and supervise repair action due to equipment failures to include:

   a. Receiving defect reports from operators and maintainers

   b. Investigating cause, nature and extent of equipment defects and failures

   c. Assessing impact of equipment failures on operations and advising command on operational limitations under various system operating modes to include:

      (1) Environmental factors (weather, sea state, etc);

      (2) Ship’s services;

      (3) Operational conditions;

      (4) Modes of operations;

      (5) Operator action at time of fault;

      (6) SOTs;

      (7) Equipment diagnostics;

      (8) Operational and maintenance history; and

(9) State of related equipment.

d. Determining requirements for equipment and repair team safety and isolating system faults

e. Identifying requirements and methods of obtaining spares to include:

(1) EMREQs;

(2) TRANREQs;

(3) High Priority Stores Requirements;

(4) PIFs;

(5) LCMM Liaison;

(6) Forward Logistics Site; and

(7) Manufacture from original drawings.

f. Planning repairs by taking command priorities and directing repair teams to include:

(1) Inputs and concerns such as:

(a) Discussing with operators and maintainers;

(b) Utilizing appropriate manuals;

(c) Determining operational considerations to rectify fault, including consultation with command and OOW/ORO/Dived Watch Leader; and

(d) Determining activities/schedules of other departments that could have an impact.

(2) Availability of spares and other resources; and

(3) Results of liaison with other departments.

g. Directing departmental personnel to rectify faults

h. Performing the duties of the departmental OPDEF coordinator to include drafting and managing OPDEFs

i. Direct repairs, liaise with shore authorities and brief command on submarine specific repair considerations that include:

(1) Starred Escape Items;

(2) First Level systems, spares and handling of OQE;

(3) License implications; and

(4) DRMIS statuses and meanings (ESMC, FIAL,TRMI, SACR, etc).

4. Teaching Points: N/A

5. Total EO Time: Throughout 12 months of training time.

6. Training and Learning Aids: Nil.

7. Remarks: Nil.

PO 012 – MANAGE BATLLE DAMAGE

EO 012.01 DESCRIBE THE COMBAT SYSTEMS ERT ORGANIZATION

1. Performance: Describe the Combat Systems Emergency Repair Team Organization

2. Conditions:

   a. Given:

      (1) CSE Dept in a ship or submarine;

      (2) References;

      (3) ERT Pack and ERT; and

      (4) Electronic ERT Pack (HFX).

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Alongside, at sea, under all conditions day or night.

3. Standard: IAW refs, the NCS Eng shall describe the ERT organization at Action, Emergency and Emergency Flying Stations. Officers in submarines shall describe the submarine damage control organization when responding to all VCEOPs, to include Action Stations. Topics to include:

   a. Composition

   b. Tools/handbooks/repairmaterial

   c. Location

   d. Ship wide resources

   e. Control

   f. Duties and responsibilities (initial responses/follow up actions)

   g. Action packs/state boards/electronic ERT pack

   h. Means of communication

i. Differences in reactions to Action Stations/Emergency Stations / Emergency Flying Stations

4. Teaching Points: N/A

5. Total EO Time: Throughout 12 months of training time.

6. Training and Learning Aids: Nil.

7. Remarks: Nil.

PO 012 – MANAGE BATLLE DAMAGE

EO 012.02 DIRECT SHIP'S ERT

1. Performance: Direct Ship's ERT

2. Conditions:

   a. Given:

      (1) CSE Dept in a ship or submarine;

      (2) References;

      (3) ERT Pack and ERT; and

      (4) Electronic ERT Pack (HFX).

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Alongside, at sea, under all conditions day or night.

3. Standard: IAW refs, the NCS Eng shall direct Ship's ERT to include:

   a. Determining status of ERT when closed up to include:

      (1) Location of personnel;

      (2) Verification of ERT equipment;

      (3) Readiness of personnel;

      (4) Communication checks;

      (5) Reporting status to command; and

      (6) Cross-checking status boards.

   b. Assessing combat systems readiness to include:

      (1) Updating equipment status;

      (2) Verifying nature of threat/situation with command; and

      (3) Reporting readiness to command.

c. Directing and briefing ERT to include:

(1) Relevant threat/situation information;

(2) Priorities of response to damage/failures;

(3) Safe routes for movement through ship;

(4) Direct action to be conducted; and

(5) Change location of ERTs as required.

d. Monitoring equipment state/damage to include:

(1) Receiving reports from ERT and CSED personnel/operators;

(2) Receiving reports from other sources (damage control/OPS);

(3) Monitoring relevant equipment alerts and alarms; and

(4) Consulting with Command to prioritize activities for action or emergency stations.

e. Directing ERT to respond to the prioritized plan considering:

(1) Casualty/emergencypower;

(2) Location of emergency/damage;

(3) Ventilation and related systems;

(4) Isolating power to various equipment; and

(5) Magazine status (fire, flood, use, adjacent compartments).

f. Officers in Submarines shall perform the following in HQ1:

(1) Advise the Incident Controller on the status of weapons, sensors, navigation systems, in addition to available work arounds and redundancy;

(2) Advise The Incident Controller on repair and restoration of weapons and electronics systems, including recommended repair priorities;

(3) Assess options for smoke clearance procedures and advise the Incident controller;

(4) Recommend, implement and monitor use of Change over Switches based on Command Priority;

(5) Perform fault correlation following Electrical Audits;

(6) Liaise with ERT, issue safety briefs, direct repair actions; and

(7) Advise the Incident Controller on HAZMAT spill procedures, milestones, risks.

4. Teaching Points: N/A

5. Total EO Time: Total: Throughout 12 months of training time.

6. Training and Learning Aids: Nil.

7. Remarks: Nil.

PO 012 – MANAGE BATLLE DAMAGE

EO 012.03 MANAGE ERT TRAINING

1. Performance: Manage ERT training

2. Conditions:

   a. Given:

      (1) CSE Dept in a ship or submarine;

      (2) References;

      (3) ERT Pack and ERT; and

      (4) Electronic ERT Pack (HFX).

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Alongside, at sea, under all conditions day or night.

3. Standard: IAW refs, the NCS Eng shall plan, conduct and assess ERT training to include:

   a. Operational team training such as:

      (1) DCOTT;

      (2) Table-top/Comms exercises; and

      (3) Force Protection.

   b. Shipboard training such as :

      (1) Emergency Stations Exercises;

      (2) Table-top/Comm exercises;

      (3) Battle problems; and

      (4) IPT.

   c. Submarine team training such as:

(1) Preparations for Ship Control and Machinery Control Training(SCT/MCT);

(2) Preparations for Integrated Platform Team Training (IPTT);

(3) At sea DC scenarios;

(4) Battle problems;

(5) Ship control training scenarios; and

(6) HQ1 training scenarios.

4. Teaching Points: N/A

5. Total EO Time: Throughout 12 months of training time.

6. Training and Learning Aids: Nil.

7. Remarks: Nil.

PO 013 – RESPOND TO NON-ROUTINE OPERATIONS

EO 013.01 DIRECT THE HAZARDOUS SPILL RESPONSE TEAM

1. Performance: Direct a Hazardous Spill Response clean-up team

2. Conditions:

   a. Given:

      (1) CSE Dept in a ship or submarine;

      (2) Class EMS manual; and

      (3) References.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Alongside, at sea, under all conditions day or night.

3. Standard: IAW refs the NCS Eng shall direct a hazardous spill clean-up team to include:

   a. Taking appropriate actions to contain the spill;

   b. Evacuating the contaminated or endangered area;

   c. Cleaning up and de-contaminating the spill site;

   d. Advising command on clean-up progress and impact;

   e. Reporting, in accordance with formation policy, the disposal action taken, probable consequences of the spill and the success of the clean-up operation;

   f. Conducting an investigation; and

   g. Officers in submarines shall in addition advise command on Otto fuel spill procedures and considerations as defined in Submarine Weapons Operating Procedures.

4. Teaching Points: N/A

5. Total EO Time: Throughout 12 months of training time.

6. Training and Learning Aids: Nil.

7. Remarks: This enabling objective is directly related to EO 004.04.

PO 013 – RESPOND TO NON-ROUTINE OPERATIONS

EO 013.02 RESPOND TO AMMUNITION EMERGENCIES

1. Performance: Respond to ammunition emergencies

2. Conditions:

    a. Given:

        (1) CSE Dept in a ship or submarine; and

        (2) References.

    b. Denied:

        (1) Nil.

    c. Environment:

        (1) Alongside, at sea, under all conditions day or night.

3. Standard: The NCS Eng shall respond to ammunition emergencies to include:

    a. High Temp alarm in GMVLS;

    b. Confirmation of Fire in GMVLS;

    c. Confirmation of spray activation;

    d. GMVLS restrained firing;

    e. Harpoon emergency;

    f. Torpedo hot run;

    g. Submarine pyrotechnic activation;

    h. Submarine Weapons Embarkation Casualty Procedures; and

    i. Participate in Submarine Weapons Embarkation Team Training.

4. Teaching Points: N/A

5. Total EO Time: Throughout 12 months of training time..

6. Training and Learning Aids: Nil.

7. Remarks: Nil.

PO 014 – MANAGE DOCKING

EO 014.01 PREPARE SHIP FOR DOCKING/UNDOCKING

1. Performance: Prepare ship for docking/undocking

2. Conditions:

   a. Given:

      (1) CSE Dept in a ship or submarine; and

      (2) References.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Alongside, at sea, under all conditions day or night.

3. Standard: IAW refs, the NCS Eng shall prepare ship for docking/undocking to include:

   a. Coordinating with other departments

   b. Coordinating with MSEO to address CSED concerns

   c. Arranging for de-ammunitioning or waiver as applicable

   d. Ensuring underwater fittings are in correct state

   e. Coordinating a departmental personnel plan

   f. Responding to docking considerations such as:

      (1) Safety/void spaces;

      (2) Hull outfits;

      (3) Tank closing up reports; and

      (4) Ship’s services.

   g. Departmental briefings

   h. Participating in and preparing for submarine licensing activities such as:

(1) Pre-Undocking Aggregate Risk Assessment;

(2) Pre-Undocking Reading of the Submarine Safety Document Register (SSDR);

(3) Manage Caveat clearance; and

(4) Undocking Certificate.

4. Teaching Points: N/A

5. Total EO Time: Throughout 12 months of training time.

6. Training and Learning Aids: Nil.

7. Remarks: Nil.

PO 014 – MANAGE DOCKING

EO 014.02 MANAGE DEPARTMENTAL ACTIONS WHILE DOCKED

1. Performance: Manage departmental actions while docked

2. Conditions:

    a. Given:

        (1) CSE Dept in a ship or submarine; and

        (2) References.

    b. Denied:

        (1) Nil.

    c. Environment:

        (1) Alongside, at sea, under all conditions day or night.

3. Standard: IAW refs, the NCS Eng shall manage departmental actions while docked to include:

    a. Conducting applicable maintenance inspection procedures for underwater fittings

    b. Ensuring correct equipment operation and maintaining restrictions

    c. Considering cold weather precautions

4. Teaching Points: N/A

5. Total EO Time: Throughout 12 months of training time.

6. Training and Learning Aids: Nil.

7. Remarks: Nil.

PO 015 – MANAGE SHIP'S TIERED READINESS PROGRAM (TRP)

EO 015.01 PROJECT MANAGE DEPARTMENTAL RESPONSIBILITIES DURING A TRP

1. Performance: Project manage departmental responsibilities during a TRP

2. Conditions:

   a. Given:

      (1) CSE Dept in a ship or submarine;

      (2) References;

      (3) Appropriate Formation and repair facility support; and

      (4) TRP Alert letter/OPORD.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Alongside, at sea, under all conditions day or night.

3. Standard: IAW refs, the NCS Eng shall project manage departmental responsibilities during a TRP by:

   a. Managing equipment readiness to meet TRP objectives to include:

      (1) Readiness inspections;

      (2) Post DWP/refit trials;

      (3) Readiness Trials, alongside and at sea;

      (4) Air Readiness Inspections (ARIs);

      (5) Ammunitioning;

      (6) Missile readiness checks WSATs;

      (7) Defect rectification/EC installation SWP;

      (8) Re-Trials;

      (9) Work-ups/Readiness Training;

(10) CTWSC; and

(11) OTT parts I & II.

b. Managing personnel to meet TRP objectives to include:

(1) Ensuring that refresher and team training are up to date; and

(2) Planning leave periods.

c. In addition, officers in submarines are to manage SUBSAFE activities throughout TRP to include:

(1) Pre Camber dive licensing activities (RRB, SSDR, Licensing) and trial requirements;

(2) Pre sea Licensing Activities (RRB, SSDR, Licensing) and trial requirements;

(3) Pre-sea Team Training requirements per Readiness State;

(4) Surfaced and dived trial program;

(5) Deep Dive trial preparations and conduct;

(6) Weapons certification, sonar and ranging trials and preparations;

(7) Level III License significance and preparations; and

(8) Team training pre-requisites for each licensing level.

4. Teaching Points: N/A

5. Total EO Time: Throughout 12 months of training time.

6. Training and Learning Aids: Nil.

7. Remarks: Nil.

PO 016 – MANAGE FORMATION LEVEL MAINTENANCE ACTIVITIES

EO 016.01 PREPARE AND STAFF FIRST LINE O&M DOCUMENTATION

1. Performance: Prepare and staff first line O&M documentation

2. Conditions:

   a. Given:

      (1) CSE Dept in a ship or submarine;

      (2) References;

      (3) Current naval maintenance management information systems; and

      (4) Appropriate Formation and repair facility support.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Alongside, at sea, under all conditions day or night.

3. Standard: IAW refs, the NCS Eng shall prepare and staff first line O&M documentation using the naval maintenance information system to include:

   a. Identifying deficiencies in maintenance schedules through the UCR process

   b. Developing new maintenance procedures and techniques, and drafting documentation for review by higher authority

   c. Evaluating new maintenance procedures and techniques

   d. Reviewing and approving regular stores demands

4. Teaching Points: N/A

5. Total EO Time: Throughout 12 months of training time.

6. Training and Learning Aids: Nil.

7. Remarks: Nil.

PO 016 – MANAGE FORMATION LEVEL MAINTENANCE ACTIVITIES

EO 016.02 LIAISE WITH FORMATION STAFF WITH RESPECT TO PRIORITY

1. Performance: Liaise with Formation staff with respect to priority

2. Conditions:

   a. Given:

      (1) CSE Dept in a ship or submarine;

      (2) References;

      (3) Current naval maintenance management information systems; and

      (4) Appropriate Formation and repair facility support.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Alongside, at sea, under all conditions day or night.

3. Standard: IAW refs, The NCS Eng shall liaise with formation staff with respect to priority to include:

   a. Identifying/referring maintenance tasks beyond ship's staff capability

   b. Prioritizing corrective maintenance, current preventive maintenance and outstanding preventive maintenance, taking into consideration:

      (1) OutstandingOPDEFs;

      (2) Ship's program;

      (3) Safety requirements;

      (4) SUBSAFE requirements;

      (5) Equipment availability; and

      (6) Resource availability.

4. Teaching Points: N/A

5. Total EO Time: Total: Throughout 12 months of training time.

6. Training and Learning Aids: Nil.

7. Remarks: Nil.

PO 016 – MANAGE FORMATION LEVEL MAINTENANCE ACTIVITIES

EO 016.03 MONITOR CONTRACT WORK

1. Performance: Monitor contract work

2. Conditions:

   a. Given:

      (1) CSE Dept in a ship or submarine;

      (2) References;

      (3) Current naval maintenance management information systems; and

      (4) Appropriate Formation and repair facility support.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Alongside, at sea, under all conditions day or night.

3. Standard: The NCS Eng shall monitor contract work to include:

   a. Contract progress againstscheduled milestones

   b. Technical compliance with contracts

   c. Reviewing contract deliverables for compliance

4. Teaching Points: N/A

5. Total EO Time: Total: Throughout 12 months of training time.

6. Training and Learning Aids: Nil.

7. Remarks: Nil.

PO 016 – MANAGE FORMATION LEVEL MAINTENANCE ACTIVITIES

EO 016.04 MANAGE TEMMIS

1. Performance: Manage Test Equipment Maintenance Management Information System (TEMMIS)

2. Conditions:

   a. Given:

      (1) CSE Dept in a ship or submarine;

      (2) References;

      (3) Current naval maintenance management information systems; and

      (4) Appropriate Formation and repair facility support.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Alongside, at sea, under all conditions day or night.

3. Standard: The NCS Eng shall manage TEMMIS to include:

   a. Reviewing TEMMIS reports; and

   b. Identifying TEMMIS requirements.

4. Teaching Points: N/A

5. Total EO Time: Total: Throughout 12 months of training time.

6. Training and Learning Aids: Nil.

7. Remarks: Nil.

PO 017 – MANAGE A SHORT WORK PERIOD (SWP)

EO 017.01 PROJECT MANAGE A SWP

1. Performance: Project Manage a SWP

2. Conditions:

    a. Given:

        (1) CSE Dept in a ship or submarine;

        (2) References; and

        (3) Appropriate Formation and repair facility support.

    b. Denied:

        (1) Nil.

    c. Environment:

        (1) Alongside, at sea, under all conditions day or night.

3. Standard: The NCS Eng shall project manage a SWP to include:

    a. Preparing for the short work period by:

        (1) Receiving SWP Alert message;

        (2) Ensuring all Work Requests (CM, PM and ECs) for SWP are entered into the naval maintenance information system;

        (3) Drafting SWP Response Message to include prioritizing work taking into consideration:

            (a) Outstanding OPDEFs;

            (b) Ship's program;

            (c) Safety requirements;

            (d) SUBSAFE requirements;

            (e) Equipment availability; and

            (f) Resource availability.

        (4) Receiving work request priority list;

(5) Sending completed work request priority list to FMF and Fleet Engineering;

(6) Attending the Ship's Contract Acceptance Meeting; and

(7) Briefing command on SWP issues.

b. Managing the short work period to include:

(1) Attending progress review meetings

(2) Addressing internal and external issues and conflicts such as:

(a) Personnel availability;

(b) Ship's activities;

(c) Equipment availability;

(d) Ship's services;

(e) Cranes;

(f) Divers;

(g) Other ships;

(h) QHM;

(i) RF work priorities; and

(j) Safety.

(3) Briefing command on the progress of SWP

(4) Demonstrate an understanding of submarine specific work period considerations

(a) Role of the Submarine Technical Officer;

(b) Defect rectification process to include vetting, DRMIS statuses;

(c) Risk Review Boards;

(d) Material Fitness Report;

(e) License Exceptions; and

(f) VCS 9+3 Operating and Maintenance Cycle.

c. Completing the short work period to include:

(1) Attending SWP Wrap-up meeting;

(2) Providing FMF feedback of overall effectiveness;

(3) Prioritizing outstanding work; and

(4) Briefing command on results of SWP.

4. Teaching Points: N/A

5. Total EO Time: Throughout 12 months of training time.

6. Training and Learning Aids: Nil.

7. Remarks: Nil.

PO 018 – MANAGE ADM(MAT) LEVEL MAINTENANCE ACTIVITIES

EO 018.01 PLAN A REFIT/DOCKING WORK PERIOD

1. Performance: Plan a refit/docking work period

2. Conditions:

   a. Given:

      (1) CSE Dept in a ship or submarine;

      (2) References;

      (3) Appropriate ADM(Mat) support; and

      (4) Appropriate Formation support.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Alongside, at sea, under all conditions day or night.

3. Standard: The NCS Eng shall plan a refit/docking work period to include:

   a. Reviewing the Refit Alert letter

   b. Reviewing the SSRWC

   c. Identifying particular defects and provide a detailed work description in the DND 650s

   d. Scheduling and completing required pre-refit surveys, trials and inspections with the assistance of formation engineering, FMF and class desk contractors in accordance with the OPSKED

   e. Submitting DND 650s to the MRSL development agent on an "as completed" basis, not awaiting the completion of all DND 650s

   f. Providing feedback to class desk on DND 650 priority list and accepted PMRS list

   g. Managing de-ammunitioning of ship

   h. Managing departmental de-storing

i. Managing departmental preparation for transit to and from contractors yard such as:

(1) Receipt of transit pack; and

(2) Personnel requirements for force protection.

j. Participating in the Supply Overhaul Assistance Program (SOAP).

k. Developing a personnel employment and training plan.

4. Teaching Points: N/A

5. Total EO Time: Throughout 12 months of training time.

6. Training and Learning Aids: Nil.

7. Remarks: Nil.

PO 018 – MANAGE ADM(MAT) LEVEL MAINTENANCE ACTIVITIES

EO 018.02 MONITOR CONTRACT WORK

1. Performance: Monitor contract work

2. Conditions:

   a. Given:

      (1) CSE Dept in a ship or submarine;

      (2) References;

      (3) Appropriate ADM(Mat) support; and

      (4) Appropriate Formation support.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Alongside, at sea, under all conditions day or night.

3. Standard: The NCS Eng shall monitor contract progress against scheduled milestones to include:

   a. Ensuring technical compliance with contracts

   b. Reviewing contract deliverables for compliance

   c. Reviewing Observation reports for defects/anomalies during refits/DWPs/Ship construction

   d. Managing set-to-work requirements to include :

      (1) Reviewing set-to-work schedule;

      (2) Providing feedback on set-to-work schedule to class desk through QAR; and

      (3) Providing technical expertise/operators as required.

   e. Ensuring system operability for undocking and acceptance to include

      (1) Completion and set-to-work of docking dependent work; and

(2) Completion and set-to-work for final acceptance.

4. Teaching Points: N/A

5. Total EO Time: Throughout 12 months of training time.

6. Training and Learning Aids: Nil.

7. Remarks: Nil.

PO 018 – MANAGE ADM(MAT) LEVEL MAINTENANCE ACTIVITIES

EO 018.03 PROVIDE SUPPORT TO QUALITY ASSURANCE REPRESENTATIVES

1. Performance: Provide support to Quality Assurance Representatives

2. Conditions:

   a. Given:

      (1) CSE Dept in a ship or submarine;

      (2) References;

      (3) Appropriate ADM(Mat) support; and

      (4) Appropriate Formation support.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Alongside, at sea, under all conditions day or night.

3. Standard: The NCS Eng shall provide support to Quality Assurance Representatives to include:

   a. Augmenting the on-site QAR team by supplying the services of certain technical and support personnel in accordance with detailed direction to be promulgated by the class desk and formation staff

   b. Ensuring qualified personnel are available to operate equipment during set-to-work period

   c. Liaising with QAR to minimize risk associated with critical milestones to include:

      (1) Government supplied materiel;

      (2) Contractor furnished materiel;

      (3) Special tools and test equipment (SPTATE);

      (4) Docking dependent work.

4. Teaching Points: N/A

5. Total EO Time: Throughout 12 months of training time.

6. Training and Learning Aids: Nil.

7. Remarks: Nil.

PO 019 – MANAGE SHIPBOARD AMMUNITION AND EXPLOSIVE

EO 019.01 DESCRIBE THE AMMUNITION ORGANIZATION

1. Performance: Describe the Ammunition Organizations

2. Conditions:

   a. Given:

      (1) CSE Dept in a ship or submarine; and

      (2) References.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Alongside, at sea, under all conditions day or night.

3. Standard: The NCS Eng shall demonstrate a detailed knowledge of the personnel involved and their roles in the management of ammunition to include:

   a. Key personnel in the shipboard ammunition organization such as

      (1) Commanding Officer;

      (2) Combat Systems Engineering Officer;

      (3) Operations Officer;

      (4) Marine Systems Engineering Officer;

      (5) Logistic Officer;

      (6) Magazine Yeoman;

      (7) Magazine Custodian;

      (8) Aircraft Crew Commander; and

      (9) Air Detachment Technician.

   b. Shore based agencies such as :

      (1) DAER;

(2) DAEME;

(3) J4 Ammo;

(4) F34;

(5) FTA;

(6) CFADs; and

(7) FMFs.

4. Teaching Points: N/A

5. Total EO Time: Throughout 12 months of training time.

6. Training and Learning Aids: Nil.

7. Remarks: Nil.

PO 019 – MANAGE SHIPBOARD AMMUNITION AND EXPLOSIVE

EO 019.02 MANAGE STORAGE AND HANDLING OF AMMUNITION

1. Performance: Manage the storage and handling of ammunition

2. Conditions:

   a. Given:

      (1) CSE Dept in a ship or submarine; and

      (2) References.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Alongside, at sea, under all conditions day or night.

3. Standard: The NCS Eng shall manage the storage and handling of ammunition to include:

   a. Considerations to be taken into account when embarking/disembarking ammunition and explosives:

      (1) Locations permitted for conducting embarking and disembarking activities;

      (2) Condition of compartments prior to embarking ammunition and explosives;

      (3) Transmitting restrictions;

      (4) Civilian personnel restrictions;

      (5) Supervision of work;

      (6) Wet weather and thunderstorm considerations;

      (7) Inspection of ammunition prior to and after Ammunitioning/ De-ammunitioning, including BIT/BITE tests of canistered missiles;

      (8) Ammunitioning/De-ammunitioning at sea;

      (9) Fire precautions; and

(10) Precautions regarding flammable and dangerous fluids.

b. Proper handling of ammunition and explosives:

(1) Safety precautions personnel must take;

(2) Use of materiel handling appliances; and

(3) Inspection and certification of materiel handling appliances.

c. Proper stowage of naval ammunition and explosives:

(1) Current regulations and marking scheme for magazines, lockers and missile canisters;

(2) Flood and spray requirements for magazines and lockers; and

(3) Security of magazines and lockers.

d. Environmental conditions to be considered when assessing the suitability of a magazine or locker:

(1) Temperature requirements;

(2) Humidity requirements;

(3) Ventilation and air conditioning requirements; and

(4) HERO and electrostatic discharge considerations.

e. Proper care of ammunition:

(1) Operations permitted in ammunition spaces and adjacent spaces;

(2) Obtaining waivers for hot work;

(3) Extraneous materials; and

(4) Prohibited articles and smoking restrictions with respect to ammunition spaces.

f. Proper action to be taken in the event of :

(1) Wetting;

(2) Corrosion;

(3) Oxidation;

(4) Temperature and humidity readings beyond limits; and

(5) Physical damage.

g. Disposal of ammunition and explosives:

(1) Jettisoning materiel and requisite messages; and

(2) Returning defective items to the depot.

h. Regulations for the inspection of:

(1) Ammunition :

(a) Normal requirements;

(b) Maintenance action required of certain ammunition, including BIT/BITE test of canistered missiles; and

(c) Verification of display and dummy ammunition.

(2) Ammunition spaces :

(a) Requirements for certification;

(b) Periodic inspections;

(c) Daily inspections;

(d) Inspections prior to docking; and

(e) Certification of ammunition spaces free from explosives.

4. Teaching Points: N/A

5. Total EO Time: Throughout 12 months of training time.

6. Training and Learning Aids: Nil.

7. Remarks: Nil.

PO 019 – MANAGE SHIPBOARD AMMUNITION AND EXPLOSIVE

EO 019.03 PERFORM AS SHIP'S AMMUNITION OFFICER

1. Performance: Perform as Ship's Ammunition Officer

2. Conditions:

   a. Given:

      (1) CSE Dept in a ship or submarine; and

      (2) References.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Alongside, at sea, under all conditions day or night.

3. Standard: The NCS Eng shall perform the duties of the ship's ammunition officer to include:

   a. Regulating ammunition holdings and expenditures for HMC ships:

      (1) Basic Load;

      (2) Baseline Ammunition;

      (3) Changes to training allocation;

      (4) Drafting requests to go below Basic Load; and

      (5) Ammunition shelf life.

   b. Managing the ammunition SLOC to include:

      (1) Responsibility for the account;

      (2) Stocktaking requirements;

      (3) Annual inspections; and

      (4) Expenditures and demands account procedures.

   c. Actioning ammunition restrictions and ammunition directives

d. Staffing reports such as :

(1) Ammunition expenditures;

(2) Verification and control of display and dummy ammunition;

(3) Report of jettisoned ammunition;

(4) Ammunition landed for inspection; and

(5) Ammunition accident/incident report.

e. Liaising with the Operations Team with respect to ammunition requirements

f. Completing required training, to wit, the Unit Explosive Safety Officer course

4. Teaching Points: N/A

5. Total EO Time: Throughout 12 months of training time.

6. Training and Learning Aids: Nil.

7. Remarks: Nil.

PO 019 – MANAGE SHIPBOARD AMMUNITION AND EXPLOSIVE

EO 019.04 MANAGE THE SHIPBOARD EXPLOSIVE SAFETY PROGRAM

1. Performance: Manage the Shipboard Explosive Safety Program

2. Conditions:

   a. Given:

      (1) CSE Dept in a ship or submarine; and

      (2) References.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Alongside, at sea, under all conditions day or night.

3. Standard: The NCS Eng shall manage the shipboard Explosive Safety Program to include:

   a. Describing explosive safety regulations;

   b. Ensuring the ship maintains all applicable explosive safety documentation; and

   c. Enforcing explosive safety regulations within the ship.

4. Teaching Points: N/A

5. Total EO Time: Throughout 12 months of training time.

6. Training and Learning Aids: Nil.

7. Remarks: Nil.

PO 020 – MANAGE DEPARTMENTAL SYSTEMS

EO 020.01 DESCRIBE ABOVE WATER SENSOR SYSTEMS

1. Performance: Describe Above Water Sensor Systems

2. Conditions:

    a. Given:

        (1) CSE Dept in a ship or submarine; and

        (2) References.

    b. Denied:

        (1) Nil.

    c. Environment:

        (1) Alongside, at sea, under all conditions day or night.

3. Standard: The NCS Eng shall demonstrate a detailed knowledge of Above Water Sensor systems to include:

    a. Purpose

    b. Capabilities, Limitations and Redundancies

    c. System diagram describing main components and signal flow

    d. Interfacing/Integration with othersystems

    e. Operating policy and procedures

    f. Modes of operation

    g. Abnormal operations

    h. Personnel and maintenance requirements

4. Teaching Points: N/A

5. Total EO Time: Throughout 12 months of training time.

6. Training and Learning Aids: Nil.

7. Remarks: Nil.

PO 020 – MANAGE DEPARTMENTAL SYSTEMS

EO 020.02 DESCRIBE COMMUNICATIONS SYSTEMS

1. Performance: Describe Communications systems

2. Conditions:

    a. Given:

        (1) CSE Dept in a ship or submarine; and

        (2) References.

    b. Denied:

        (1) Nil.

    c. Environment:

        (1) Alongside, at sea, under all conditions day or night.

3. Standard: The NCS Eng shall demonstrate a detailed knowledge of Communication systems to include:

    a. Purpose

    b. Capabilities, Limitations and Redundancies

    c. System diagram describing main components and signal flow

    d. Interfacing/Integration with other systems

    e. Operating policy and procedures

    f. Modes of operation

    g. Abnormal operations

    h. Personnel and maintenance requirements

4. Teaching Points: N/A

5. Total EO Time: Throughout 12 months of training time.

6. Training and Learning Aids: Nil.

7. Remarks: Nil.

PO 020 – MANAGE DEPARTMENTAL SYSTEMS

EO 020.03 DESCRIBE COMMAND AND CONTROL SYSTEMS

1. Performance: Describe Command and Control systems

2. Conditions:

   a. Given:

      (1) CSE Dept in a ship or submarine; and

      (2) References.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Alongside, at sea, under all conditions day or night.

3. Standard: The NCS Eng shall demonstrate a detailed knowledge of Command and Control systems to include:

   a. Purpose

   b. Capabilities, Limitations and Redundancies

   c. System diagram describing main components and signal flow

   d. Interfacing/Integration with othersystems

   e. Operating policy and procedures

   f. Modes of operation

   g. Abnormal operations

   h. Personnel and maintenance requirements

4. Teaching Points: N/A

5. Total EO Time: Throughout 12 months of training time.

6. Training and Learning Aids: Nil.

7. Remarks: Nil.

PO 020 – MANAGE DEPARTMENTAL SYSTEMS

EO 020.04 DESCRIBE UNDERWATER SENSOR SYSTEMS

1. Performance: Describe Underwater Sensor systems

2. Conditions:

   a. Given:

      (1) CSE Dept in a ship or submarine; and

      (2) References.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Alongside, at sea, under all conditions day or night.

3. Standard: The NCS Eng shall demonstrate a detailed knowledge of Underwater Sensor systems to include:

   a. Purpose

   b. Capabilities, Limitations and Redundancies

   c. System diagram describing main components and signal flow

   d. Interfacing/Integration with other systems

   e. Operating policy and procedures

   f. Modes of operation

   g. Abnormal operations

   h. Personnel and maintenance requirements

4. Teaching Points: N/A

5. Total EO Time: Throughout 12 months of training time.

6. Training and Learning Aids: Nil.

7. Remarks: Nil.

PO 020 – MANAGE DEPARTMENTAL SYSTEMS

EO 020.05 DESCRIBE ELECTRONIC WARFARE SYSTEMS

1. Performance: Describe Electronic Warfare systems

2. Conditions:

   a. Given:

      (1) CSE Dept in a ship or submarine; and

      (2) References.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Alongside, at sea, under all conditions day or night.

3. Standard: The NCS Eng shall demonstrate a detailed knowledge of Electronic Warfare systems to include:

   a. Purpose

   b. Capabilities, Limitations and Redundancies

   c. System diagram describing main components and signal flow

   d. Interfacing/Integration with othersystems

   e. Operating policy and procedures

   f. Modes of operation

   g. Abnormal operations

   h. Personnel and maintenance requirements

4. Teaching Points: N/A

5. Total EO Time: Throughout 12 months of training time.

6. Training and Learning Aids: Nil.

7. Remarks: Nil.

PO 020 – MANAGE DEPARTMENTAL SYSTEMS

EO 020.06 DESCRIBE DATA PROCESSING SYSTEMS

1. Performance: Describe Data Processing systems

2. Conditions:

   a. Given:

      (1) CSE Dept in a ship or submarine; and

      (2) References.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Alongside, at sea, under all conditions day or night.

3. Standard: The NCS Eng shall demonstrate a detailed knowledge of Data Processing systems to include:

   a. Purpose

   b. Capabilities, Limitations and Redundancies

   c. System diagram describing main components and signal flow

   d. Interfacing/Integration with othersystems

   e. Operating policy and procedures

   f. Modes of operation

   g. Abnormal operations

   h. Personnel and maintenance requirements

4. Teaching Points: N/A

5. Total EO Time: Throughout 12 months of training time.

6. Training and Learning Aids: Nil.

7. Remarks: Nil.

PO 020 – MANAGE DEPARTMENTAL SYSTEMS

EO 020.07 DESCRIBE NAVIGATION SYSTEMS

1. Performance: Describe Navigation systems

2. Conditions:

   a. Given:

      (1) CSE Dept in a ship or submarine; and

      (2) References.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Alongside, at sea, under all conditions day or night.

3. Standard: The NCS Eng shall demonstrate a detailed knowledge of Navigation systems to include:

   a. Purpose

   b. Capabilities, Limitations and Redundancies

   c. System diagram describing main components and signal flow

   d. Interfacing/Integration with othersystems

   e. Operating policy and procedures

   f. Modes of operation

   g. Abnormal operations

   h. Personnel and maintenance requirements

4. Teaching Points: N/A

5. Total EO Time: Throughout 12 months of training time.

6. Training and Learning Aids: Nil.

7. Remarks: Nil.

PO 020 – MANAGE DEPARTMENTAL SYSTEMS

EO 020.08 DESCRIBE WEAPON SYSTEMS

1. Performance: Describe Weapon systems

2. Conditions:

   a. Given:

      (1) CSE Dept in a ship or submarine; and

      (2) References.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Alongside, at sea, under all conditions day or night.

3. Standard: The NCS Eng shall demonstrate a detailed knowledge of Weapon systems to include:

   a. Purpose

   b. Capabilities, Limitations and Redundancies

   c. System diagram describing main components and signal flow

   d. Interfacing/Integration with other systems

   e. Operating policy and procedures

   f. Modes of operation

   g. Abnormal operations

   h. Personnel and maintenance requirements

4. Teaching Points: N/A

5. Total EO Time: Throughout 12 months of training time.

6. Training and Learning Aids: Nil.

7. Remarks: Nil.

PO 020 – MANAGE DEPARTMENTAL SYSTEMS

EO 020.09 PERFORM THE DUTIES OF THE CSEO

1. Performance: Perform the duties of the CSEO

2. Conditions:

   a. Given:

      (1) CSE Dept in a ship or submarine;

      (2) References; and

      (3) Supervision.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Alongside, at sea, under all conditions day or night.

3. Standard: The NCS Eng shall perform as the CSEO to include:

   a. Providing written and oral orders required to direct the safe and efficient operation of combat systems

   b. Monitoring the start up, running, and shutdown of equipment and systems to include:

      (1) Observing and conferring with on-watch personnel and Combat Systems Section Heads;

      (2) Reviewing system logs and records;

      (3) Reviewing pre-sail and periodic equipment checks;

      (4) Conducting rounds; and

      (5) Initiating defect rectification by :

         (a) Identifying system deficiencies; and

         (b) Analyzing the effects of equipment degradation and assessing impact on other systems and ship's operations.

c. Monitoring the capability of departmental personnel for safe and efficient operation of systems

d. Managing technical documentation

e. Advising command with respect to:

(1) Availability;

(2) Operating Modes;

(3) Capabilities and limitations;

(4) Degradation/Re-configuration;

(5) Repair requirements;

(6) Personnel;

(7) Safety;

(8) The situation;

(9) Possible causes;

(10) Engineering and operational impact;

(11) Options to deal with the situation with details such as:

(a) Work arounds;

(b) Repair way ahead;

(c) Operational impact and limitations;

(d) Supply considerations; and

(e) Timeline.

(12) Recommendation;

(13) Implementation plan;

(14) Subject matter experts and external agencies that you would liaise with or require support from; and

(15) Administration and documentation requirements.

4. Teaching Points: N/A

5. Total EO Time: Throughout 12 months of training time

6. Training and Learning Aids: Nil

7. Remarks: Nil

QUALIFICATION STANDARD ANNEXES

ANNEX A – MAIN REFERENCES

1. This annex lists all references used in this QS and TP.

2. This is to be completed fully, for a tempory list please find the list below the table.

REF #	NDID/DOCUMENT CONTROL NO	TITLE OF REFERENCE
A		**Official CAF Publications and Directives that relate to this Qualification**
A01		
A02		
A03		
A04		
B		**(Non-CAF) Government of Canada Publications & Foreign Military (Non-DND) Publications that relate to this Qualification**
B01		
B02		
B03		
C		**Civilian Publications and Manuals that relate to this Qualification**
C01		
C02		
D		**Other Publications and Manuals that relate to this Qualification**
D01		
D02		

NOTE: If additional references are identified during the TPWB this annex can be updated secretarially through NPTG HQ.

References include, but are not limited to:

DAODs, QR&Os

Ships Standing Orders

Guide to the Divisional System

NAVORDS, MARLANTORDS, MARPACORDS

NPTORDS

ADM(MAT) Policies:

- C-03-005-033/AA-000 – Naval Engineering Manual (NEM)
- C-03-005-012/AM-001 – Naval Material Management System Manual (NaMMs)
- C-23-005-000/AG-001 – Naval Material Regulation for Surface Ships (NMRSS)

CFCDs:

- 102 Combat Readiness Requirements
- 109 Naval Signature Management Manual
- 129 Readiness and Sustainment Policy
- 132 RCN Damage Control Manual
- 133 RCN Emergency Repair Team Manual
- 122 VIC Weapons Certification Policy
- 103 Submarine Escape Manual

Readiness Training Guide

Explosive Safety Manual, Volume II

Class-specific Safety and Environment Management System (SEMS)

CF Mil Pers Instructions (CF MPIs)

Canadian Forces Administrative Orders (CFAOs)

NEMTIs

C-12-148-A00/MB-001 – Shipborne Helicopter Operating Procedures (SHOPs)

National Defence Security Orders and Directives (NDSODs)

A-PD-055-002/PP-00111511 – Occupation Specifications: Naval Combat Systems Engineering Officer

C-23-VIC-000 Series of CFTOs

NAVORD 1150 SUBSAFE policy

Submarine Technical Instructions (SMTI)

Victoria Special Technical Instructions (VSTI)

Victoria Class Emergency Operating Procedures (VCEOPs)

Victoria Class Standard Operating Procedures (VCSOPs)

Victoria Class Submarine Standing Orders (VCSSOs)

Submarine Books of Reference (BRF)

Submarine Sea Training Submarine Work up Guide

Submarine Readiness Training guide

Damage control Manual Volume 4

ANNEX B – TASK LIST

1. The task list includes all of the tasks associated with this qualification, and the outcome of the Train / No-train decisions. If a task is designated as requiring training, the PO where it is covered is indicated. If a task is designated as not requiring training, the justification for that designation is provided.

2. The following table defines some of the criteria used to support a no-train designation in the task list. Other reasons may be used if required.

CRITERIA	DEFINITION
A	**Previously qualified.** Tasks and knowledge already possessed by the student by CAF and/or RCN courses, training and/or experience.
B	**Will receive on future courses.** Tasks and knowledge will be covered as part of other CAF and/or RCN courses and/or training.
C	**Difficulty.** Task is not difficult to learn; formal training is not required.
D	**Importance.** Task is not important; it has no significant impact on mission success.
E	**Frequency.** Task is rarely performed, no formal training is required.
F	**Lack of equipment.** Currently there is no equipment in the CAF inventory to train this task.

Task #	Task Statement	Train / No Train	PO #	Substantiation/Comments

3. This table will be completed at a later date. To find the master task list please refer to the occupation specification found at https://sm-gdd.forces.mil.ca/en-CA/Occupation/Details/3394

ANNEX C – QS SCALAR

1. Manage Departmental Personnel

BT5	BT15	CT55	CT50
BT10			CT10
			CT15
			ET15

2. Liaise with laboratories/industries/establishments to support Naval Engineering

AT30

3. Manage Departmental Financial Resources

MT50
MT40

4. Manage Shipboard Risk

AT40	FT45	FT35	FT5	FT50
	FT40			

5. Prepare Technical Documentation

AT5	DT105	GT15

6. Participate in Interdepartmental Operational Planning

GT70	DT125	DT135

7. Manage Signature Management Program

IT20	DT120
DT55	

8. Manage Shipboard Configuration

HT35
LT40
LT55

9. Manage Tests and Trials Program

KT40
KT20
KT10
KT5

10. Manage Readiness

DT130B(Refer to ROP)		DT130		DT10	DT15
DT35					
DT50	DT5	DT65	DT70		

11. Manage Defect Rectification

DT40
DT95
IT10
DT115
GT75
DT20
DT45
GT35

12. Manage Battle Damage

DT75
ET20
ET10

13. Respond to Non-Routine Operations

ET30 CS	ET5 CS	

14. Manage Docking Work Period

IT15

15. Manage Ship's Tiered Readiness Program:

DT(TBD)

16. Manage Formation Level Maintenance Activities

HT40		LT80	HT30	GT100
LT75	GT40	DT110	HT20	
	GT45	DT115	HT25	

17. Manage a Short Work Period

IT (TBD)

18. Manage ADM(Mat) Level Maintenance Activities

JT40	MT20
JT5	MT30

	JT30		NT120	MT25
	NT150	MT35	CONNECTED TO	
	NT155			

19. Manage Shipboard Ammunition and Explosive Safety

FT15

20. Manage Systems

OT40	OT45	OT50	OT55	OT60	OT65	OT70	OT75
AT80	GT55	GT55	GT55	GT55	GT55	GT55	GT55
GT55	GT60	GT60	GT60	GT60	GT60	GT60	GT60
GT60	GT50	GT50	GT50	GT50	GT50	GT50	GT50
GT50	GT65	GT65	GT65	GT65	GT65	GT65	GT65
GT65							

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ANNEX D – RECORD OF DECISIONS QSWB

1. This record summarizes the conduct of the QSWB including major activities, important discussions and contentious issues, as well as all decisions and recommendations made by the Board. A description of the reasoning which supports a decision or recommendation is provided when required.

Title	Naval Combat Systems Engineering Assistant Head of Department Qualification Standard and Plan			
Location of Board	CFNES OT Div			
Dates	8 Feb to 25 Feb 05			
Board Composition				
	Surname	Initials	Rank	Unit
Chair	Stuart	R.A	LCdr	CFNES
SME	Harrison	S.C.W	LCdr	CFNES
SME	Bradford	T.L.	LCdr	CANFLTPAC
SME	Gregg	S.C	LCdr	MOG5HQ
DMTE Rep	Shirriff	T.	LCdr	DMTE
SME	Budden	H.L.	Lt(N)	HMCS IROQUOIS
SME TDO	Higdon	T.S.W	Lt(N)	NDQAR
Secretary	Checketts	A.	Lt(N)	DMTE
NTDSS Operator	Waller	R.D	Lt(N)	FMF CB
	Jones	R.	LS	CFFS(E)

INTRODUCTION

1. The Chairman opened the proceedings by welcoming everyone to the QSPWB and provided a brief outline of the objective, scope, constraints and process for the QSPWB. The QSPWB members briefly introduced themselves and Lt(N) Checketts presented a detailed overview of CFITES and the QSP process.

2. It was assumed that the candidates would arrive for the A/HOD training having completed their Basic Occupation Qualification (BOQ) and would therefore be at the requisite skill level to perform the tasks expected of the A/MSEO or A/CSEO job in the respective Integrated Occupation Specification.

3. The QSPWB was tasked to develop the HOD training for both NCS and MS Eng officers. The initial focus of the QSPWB was to examine the common tasks between both occupations and to group them in order to maximize efficiency. This involved a detailed review of the Integrated Occupation Specifications (references A and B). This review resulted in a list of recommended changes to both Integrated Occupation Specifications. These recommended changes can be categorized in three areas:

a. Wording changes/Incompatibilities between MS Eng and NCS Eng (Annex A);

b. Skill level changes related to specific tasks (Annex B); and

c. Knowledge level changes (Annex C)

4. Due to the nature of the recommended changes to the Integrated Occupation Specifications and on the advice of the DMTE representative, the QSPWB assumed that the recommended changes to the Integrated Occupation Specifications would be accepted. This meant that the development of the HOD training used the tasks, skills and knowledges from references A and B in addition to the changes recommended at the Annexes as the baseline.

TRAIN/NO TRAIN DECISIONS

5. As noted above, it was assumed that the candidates arriving to carry out the A/HOD training would have the requisite skill level associated with the A/MSEO or A/CSEO job in the respective Integrated Occupation Specification. This was important as the QSPWB focussed only on tasks where there was a skill level increase for the task from the assistant job (A/MSEO or A/CSEO) to the HOD job (MSEO or CSEO). The following items were determined to be NO TRAIN tasks:

a. FT30 MS Perform as Unit/Command General Safety Officer - As noted in Annex A, this item is only in the MS Eng IOS. It was recommended that this task be removed from the MS Eng IOS as it is not specifically related to the job of an MS Eng, rather it is a task that could be assigned to any qualified person within a unit. Further, there is specific UGSO training given elsewhere;

b. Duty Area O – Marine/Combat Systems – As noted in Annex A, it is recommended that each task in the MS Eng IOS be split into two tasks, “Manage” and “Assess” to mirror the NCS Eng IOS. The “Assess” tasks are no train since there is no increase in skill level from the A/MSEO or A/CSEO job to the MSEO or CSEO job, recognizing the fact that the MS or NCS Eng will have achieved a level 3 for the “Assess” tasks at completion of the BOQ (i.e. phase VI);

ISSUES DISCUSSED

6. The QSPWB discussed many issues in the development of the QSP. Many of the issues related to the Integrated Occupation Specifications and are captured in Annexes A to C. There were other issues that did not fall directly into these categories. These issues are captured below and, where necessary, present the reasons that a particular decision/recommendation was made on the issue.

a. Conducting A/HOD training on a different class of vessel than BOQ – There were two issues related to this point –what level of systems knowledge is expected of the A/HOD and should additional time be given to a trainee that does change platforms after BOQ:

   i. Level of systems knowledge – It was agreed by the QSPWB that the Manage Marine/Combat Systems Performance Objective (PO 020 for NCS Eng and PO 021 for MS Eng) addressed the expected system level knowledge for the A/HOD, and
   ii. Additional time – It was agreed that no additional time would be automatically given to a trainee that changes class of vessel after BOQ. This would be addressed in Chapter 1, specifically stating that the CO has the authority to grant a 25% time extension (in accordance with MARCORD 9-28) and, a change of class of vessel after BOQ is a valid reason to grant this extension;

b. The QSPWB agreed that the Naval Technical Manager Refresher (formerly TORC) is a useful course for the A/HOD to attend prior to joining their ship as the A/HOD;

c. Monitoring of training progression – The discussion centered on the issue of who should be monitoring progression. The overall responsibility for this lies with DMTE. Some ideas that were presented included having CANFLT involved (i.e. GTO / F4 Eng) or having CFNES / CFFSE involved. The QSPWB recommended that DMTE retain the overall responsibility for monitoring trainee progression with the assistance of formation agencies. The specifics of who will perform this on each coast would have to be worked out appropriately. The QSPWB recommended that, given the fact that the ships report to the Commander of the CANFLT organizations (CCFL/CCFP), the GTO / F4 Eng would be the appropriate coastal OPI;

d. NBCO for NCS Eng submariners – It was determined that this topic was outside the scope of the QSPWB;

e. Divisional Officer Duties – It was noted that in the BOQ OJPRs, there is a PO to “Lead Personnel” that is common to the MS and NCS Eng, however, there is not a specific EO in the afloat phase to act as a divisional officer. The OJPRs state in the instructions to the MSEO/CSEO that the candidate (i.e. phase VI) shall be given divisional officer duties, but this is not reflected in the OJPR. It was recommended that an afloat EO be added to the QSPs for the BOQs;

f. Manage Readiness States – A lengthy discussion on this topic was held. The decision was made to break this into three elements:

   i. Manage Readiness Alongside,
   ii. Manage Readiness At Sea, and
   iii. Tiered Readiness Program;

g. Manage Marine/Combat Systems – Managing the marine/combat systems was determined to be an all encompassing requirement – all the other performance objectives provided supporting tasks, skills and knowledges to performing as the head of the department. Adding a "Perform the duties of the CSEO/MSEO" Enabling Objective to each of the Manage Systems Performance Objectives emphasized this;

h. It was recommended that, prior to completing the HOD qualification, the A/CSEO complete the Unit RadHaz Officer course and the A/MSEO complete the Unit Environmental Officer course;

i. Risk Analysis - Knowledge of how to conduct risk analysis needs to be increased further through formal training. The QSPWB recommended that formalized lectures on the current risk analysis processes and policies such as SUBSAFE be added to the BOQ ashore coursing. In addition, the knowledges associated with this item (K0745 NCS / K0795 MS) should be increased to 2 for the A/MSEO / A/CSEO. At the MSEO / CSEO level these knowledges should increase to 3 and continue at 3 for post HOD positions;

j. Reference CD – There was a discussion on the merits of providing a reference CD for the A/HOD. It was decided that providing a CD (or other media) with the references for the AHOD was not beneficial for the training. There were two reasons for this decision. The first was that the QSPWB agreed that the ability to find, interpret and understand the references is an important skill to learn and secondly, the references change on a regular basis;

k. Number of Trainees embarked – The AHOD position is an established billet with a valid job – that of being the Assistant HOD. It should be noted that this position comes with the responsibility to carry out OJT while performing the duties required of the Assistant HOD. The trainee completing their BOQ (phase VI) is there primarily to carry out their training. If there is no AHOD, the phase VI trainee should not be used primarily to fill the gap, as it will impact on their training. It was recommended that no more than one A/CSEO and one A/MSEO be posted to each ship/submarine. It was acknowledged that there may be situations where an overlap occurs,

however, the amount of time for the overlap should be minimized;

l. MS Eng BOQ, EO 403.2 – It was noted that this enabling objective has no mention of the PROTECTEUR Class propulsion control system. This needs to be examined and if phase VI candidates continue to be sent to PROTECTEUR Class ships, suitable amendments to the MS Eng OJPR should be made;

m. Guidance to the ship (CO/XO/HOD) for A/HOD training –This point was discussed and consideration was given to:

   i. Having the CCA send a letter to the ship with specific guidance for the HOD and a "contract" to be signed by the HOD, acknowledging their role; or
   ii. Having a coastal representative (i.e. CFNES/CFFSE) visit the ship when an AHOD is posted in to clarify the training requirements and address any questions of concerns that the ship may have; or
   iii. Inclusion of the specific guidance for the HOD training in the "DMTE Guidance to COs" annual letter.

   This guidance will be included in the front of the OJT package and it is recommended that it be included in the DMTE drafted "Commanding Officers Training Responsibilities" annual letter;

n. NCS Eng BOQ, EO 411.4, para 10.b. - Recommend this be re-written to indicate that the trainee will participate in HAZMAT cleanup in various positions (ie. Member of Cleanup team, dressing area assistant, OSC, command plotter);

o. In the NCS Eng BOQ OJPR, EO 414.5, para 10c ("Plan and conduct a DMP/DWP/SWP") should be deleted as this is an AHOD requirement. Note that this may be a transcription error, as this statement is not in the equivalent MS Eng BOQ EO (EO 13.5); and

p. It is recommended that the theory of heavy lift ships should be added to the MS Eng BOQ (EO 10.1).

7. There was a lengthy discussion on assessment. Historically, the HOD qualification has been assessed through an oral board after completion of the OJT package. The QSPWB went through the process of determining how to train these qualifications and it was determined that the best method was through OJT. Within the chapter 4 that was developed by the QSPWB, which forms the core material for the OJT package, each of the enabling objectives has a "Test Details" section that dictates what the candidate must do to complete a particular enabling objective.

8. It was further determined that completion of an oral board after the OJT package has been completed is the best method of formally assessing a candidate. Having a board with the content controlled by the CCA and the board conducted by CFNES for MARLANT candidates and by CFFSE for MARPAC candidates will ensure that a common standard is applied to all trainees.

OUTSTANDING ACTION REQUIRED

9. The core content of the QSP was completed by the QSPWB. The CCA now has all of this material and is progressing the remaining administrative work to finalize the QSP.

RECOMMENDED WAY AHEAD

10. The completed QSP will be forwarded to DMTE for approval of chapters 1 and 2. Once approved, CFNES as CCA will approve the remainder of the QSP. It is anticipated that the revised training package for both MS and NCS Eng AHODs can be ready to be distributed to the trainees commencing in APS 05.

CONCLUSION

11. I consider this QSPWB to have been a success. The team was able to maintain focus for the three week period. The QSPs created will not result in a radical change in A/HOD training, however, the NTO community will have a current and, in the opinion of the QSPWB members, a valid, relevant and more applicable guide to training our future heads of department.

// original signed by //
R.A. Stuart

Lieutenant Commander

Chairman, Combined MS and NCS Eng HOD QSPWB

Annex A Integrated Occupation Specification - Wording Changes/Incompatibilities

Annex B Integrated Occupation Specification – Skill Value Changes

Annex C Integrated Occupation Specification - Knowledge Value Changes

Annex A - Integrated Occupation Specifications - Wording Changes/Incompatibilities

- BT005 – Reword from “manning shortfalls” to “Personnel Shortages”
- BT NEW: “Counsel Performance” should be added to both IOS
- CT10 – MS wording should mimic NCS – i.e. add “assess”
- CT15 MS – wording should mimic CT 15 NCS – i.e. “Plan/conduct/assess …”
- DT15 – Reword to include “Operational” Degrees of Readiness
- DT NEW – Create a new requirement titled “Advise on Ship’s Tiered Readiness Program”
- DT35: Reword from “Advise on technical readiness” to “Advise on mission readiness”
- DT40 MS should be added to NCS and DT40 NCS should be added to MS
- DT65/70 MS needs to be added to NCS
- MS Staff O needs to know about NCS systems but the reciprocal for NCS is not necessary (CT35, DT 35 MS)
- Insert DT5 MS into NCS
- DT65 “Assess pre-sail checks” not included in MS. It needs to be added.
- Amend DT115 NCS to read “Identify requirements for and methods of obtaining spare parts”. For the MS IOS, insert this statement to replace DT120 MS.
- There is no “Participate in planning ship’s ops/programme/routine” (DT125 NCS) for MS. This should be added to the MS IOS.
- DT130 NCS – This task will be divided into two separate tasks: “Maximize combat systems availability within the stated notice for power” and “Maximize combat systems availability at sea”. These two tasks also need to be inserted in the MS IOS
- DT135 NCS – a similar task should be created for MS. The QSPWB decided on “ Assess the impact of MSE systems on Combat Systems”
- ET 5 NCS - “Respond to missile emergencies” is not applicable to all classes of ship. The task should be reworded to “Respond to ammunition emergencies”
- ET 30 MS should be reworded to match ET 20 NCS
- ET35 MS can be deleted because it is encompassed in DT45
- There should be a “Direct fuel spill response team” in the MS IOS to mimic ET30 NCS “Direct HAZMAT Cleanup team”. This task was added as ET35 MS.
- FT5 MS – Reword from “Prepare” to “Ensure compliance”
- FT5 – The task should be reworded to match the EMS SOP to read “governing procurement, stowage, handling, disposal and training of hazardous materials”
- FT30 MS - Performing as the UGSO is listed as an MS task. UGSO is not an occupation specific job, therefore it should not be included in the IOS. It should be replaced with FT40 NCS “Ensure compliance with general safety policy and procedures”.
- GT50 – Change the wording from “Conduct” to “Monitor”
- GT95 NCS should be inserted in the MS IOS, although the task is not applicable for this QSP

- GT NEW – “Participate in the Test Equipment Maintenance Management Information System (TEMMIS) program” should be added to the MS IOS
- IT NEW - “Project Manage a Short Work Period” should be added to both IOS
- Adjust the wording of JT5 MS to match JT5 NCS. Remove “Short Work Period” from the description because a new task will be created to focus solely on SWPs
- JT25 MS does not appear in the NCS IOS and should be added. However, this task is not applicable to this QSP
- Duty Area K should be changed in NCS to mimic the MS IOS.
- LT 75 MS is absent from the NCS IOS and needs to be added
- LT 85 NCS is not present in the MS IOS and should be added. However, this task is not applicable to this QSP
- MT40 NCS needs to be added to the MS IOS
- MT45 NCS and MT50 NCS are absent from the MS IOS and should be added.
  The QSPWB also recommends that these two tasks belong in Duty Area A
- Duty Area N differs between the NCS and MS IOS. The tasks listed in the MS IOS are more applicable and should be copied to the NCS IOS with the exception of NT100, which should not be a task for the Asst/EO
- In section O in the MS IOS, each task includes “Manage/Analyze”. The format is different in the NCS IOS where the two words are separated into different tasks. Since Analyzing systems does not result in an increase in knowledge, the section O tasks for MS should mirror the format of the tasks in the NCS IOS
- OT55 and OT60 NCS should be reworded from “Under/Above Water Warfare systems” to “Under/Above Water Sensor systems”
- OT80 NCS – The mirror task “Manage/assess Combat Systems” should be included in the MS IOS. This task is trained at the NEI level, so it is not applicable to this QSP
- OT140 MS – The task should be reworded to remove “Main” from “Main Auxiliary Systems”
- RT25 NCS – This task should be removed because it is not appropriate to have the Asst/CSEO involved in Ship/Design/Construction
- K1065 MS/ K1125 NCS – The knowledge should be reworded from “there” to “their”

Annex B - Integrated Occupation Specifications - Skill Value Changes

- AT40 A/MSEO should be a 2
- AT45 A/MSEO should be a 2
- DT40 A/MSEO, Jnr Eng and Jnr Eng SO should be a 2 to match the NCS IOS
- KT25 should be a 2 for the Assistant and the HOD – when deployed, you may have the responsibility of witnessing contractor trials
- LT40 MS LCMM is listed as a 2 – it should be changed to a 3 to mimic the NCS IOS
- LT 60 MS MSEO is listed as a 3 and should be changed to a blank
- LT 80 in both MS and NCS for the Assistant, Jr Proj Eng and Jr Eng SO should be a 2
- MT25 and MT30 MS Assistant is rated a 3 and should be a 2.
- Section M LCMM scoring is not compatible between NCS and MS. The scoring should reflect the NCS IOS except MT35, LCMM should be a 3 vice a 2
- MT 45 should be blanked for Assistant and HOD since on board ship, the HOD is not involved in the CP process, only in managing a budget, which is covered under MT50 CS. The QSPWB recommends that MT 45 change from a 2 to a 3 at the Jr Eng SO level. The basics of CP are taught in BOQ ashore (phase VI), achieving a level 2.

Annex C - Integrated Occupation Specifications - Knowledge Value Changes

- K0235 MS – Scoring should be a 2 for all positions to reflect the NCS scores
- Remove K0495 NCS/ K0540 MS
- K0680 NCS/ K0730 MS should increase to a 2 for A/CSEO and up and should become a 3 for the Snr Eng. Reword from "Submarine safe-to-dive" to "SUBSAFE" policy.
- Add a knowledge item for Submarine Materiel Certification. The scoring should be Level 2 until Snr Eng, where it is then increased to a 3
- Note that the above 2 items would impact the BOQ qualification
- K1205 – CSEO knowledge level should be a 3 while the Jr Eng, Jr Eng SO and LCMM should be a 2
- K1225 –Change scoring to a 1 for Asst and CSEO/MSEO
- K1230 – Scoring for the different positions should be re-examined
- K0720 MS – Recommend that the value be increased to a 3 for MSEO
- K0825 MS – Recommend that the value be increased to a 3 for MSEO
- K0810 MS – Recommend that the value be increased to a 2 for Asst, 3 for MSEO
- K0880 MS - Recommend that the value be increased to a 3 for MSEO
- K1055, K1060 MS – Recommend that the value be increased to a 2 for all groups
- K1055, K1060, K1070, K1075 MS/K1115-1135 NCS need their scores examined and where/when this knowledge is increased.
- K2005 MS – Recommend that the value be decreased to a 1 for Asst and MSEO
- K2010-K2085 MS: Recommend that the value be increased to a 3 for Asst
- K2090 MS – This knowledge is a duplicate of K0035, it should be deleted
- K2515, K2520, K2535, K2540 MS – There is no equivalent knowledge in the NCS IOS, it needs to be added
- K2525, K2530, K2545 MS – These should be removed because they are already being covered under specific knowledge areas
- K2555, K2560 MS – These should be removed because it is referencing the knowledge of a form
- K2565-2590 MS – Recommend that the value be increased to a 3 for Asst
- K2595, K2615 MS - There is no equivalent knowledge in the NCS IOS, it needs to be added
- K2620 MS - Recommend that the value be increased to a 3 for Asst
- K2630 MS - Recommend that the value be increased to a 3 for Asst. Also, it should be added to the NCS IOS
- K2635-2650 MS – Recommend deletion since they are OOD qualifications and not part of this phase of training
- K2655-2665 MS - No equivalent knowledge in the NCS IOS, needs to be added
- K2670 MS – The wording is vague, it should be reworded and then added to the NCS IOS
- K2675 MS - There is no equivalent knowledge in the NCS IOS, needs to be

added

- K2680 MS – Recommend that the value be decreased to a 2 for MSEO and add the knowledge to the NCS IOS
- K2685-2705 MS - There is no equivalent knowledge in the NCS IOS, needs to be added

RECORD OF AMENDMENT & AMPLIFICATION

1. The record of amendment and amplification will be filled in whenever a change to the QS is initiated. This section is used to provide amplification details and provide a history of the changes made. In the requirement for the amendment section, the member inputting the changes will transpose the information from the submitted QS Change Request (QSCR), thus ensuring the QSP has the most relevant information.

Version:	3.1
Date: 25 May 2023	**Who Initiated Change:** HMCS Venture
Requirement for the Amendment: Rewrite of the QSP into the new format. No changes to content were made	
Description of Changes: New format of the QSP, no changes to content	
Change Tracking number:	
Trg Mgr: Lt(N) Newcombe	

TRAINING PLAN ANNEXES

ANNEX E – RESOURCES, EQUIPMENT, SUPPLIES, AND ANCILLARIES

1. There are no resources, equipment, supplies, and other elements required to support this training.

ANNEX F – COSTING INFORMATION

This course has no cost as it is an OJPR to complete while posted to a sea-going vessel as the assistant head of department.

ANNEX G – SUGGESTED SEQUENCING AND TIMINGS

1. This course will take and estimated 365 days to complete the OJPR and complete the final board.

ANNEX H – ASSESSMENT PLAN

EC / PC	Scope	Purpose	How	When	Time	Resources	Limitations	Pass/Fail Criteria	Impact
Entire QSP	All POs	To assess trainee skills and knowledge to stand as a naval combat systems engineering head of department	Board-style assessment	Upon completion of OJPR, and 365 days as assistant head of department	Indicate if the time identified is the amount of time per student or the total time to complete the EC	Board room at local TE	Nil	See annex I	Failure will require TRB with extension and reboard if deemed necessary

ANNEX I – ASSESSMENT CHECKLISTS

PART 1 – EO 012.02 DIRECT SHIP'S ERT

1. **Purpose.** Directing the Ship's Emergency Response Team (ERT) fighting organization is a critical part of the CSEO position and needs to be assessed thoroughly and realistically to understand the candidate's ability to act in shipboard conditions. This requirement is captured in OJPR EO 012.02 Direct Ship's ERT.

2. **Conditions.** The candidate should be afforded similar conditions and pressures inherent in realistic Battle Damage management situations. This includes availability of job aids, communication systems, reporting mechanisms, supporting staff, and realistic faults with shipboard impacts.

3. **Expectations.** The candidate shall be provided with a scenario involving ERT. To successfully challenge this part of the board, the candidate is expected to demonstrate understanding of the threat environment and assess combat systems readiness with respect to Command Priorities. The candidate should realistically respond to the situation including:

    a. Immediate actions;
    b. Directing the ERT organization;
    c. Developing and maintaining the combat systems picture;
    d. Completing at least two Command Conferences;
    e. Completing at least two Power Isolation loops;
    f. Successfully correlating system faults;
    g. Prioritizing repairs according to Command Priority;
    h. Passing information to and from Command; and
    i. Offering recommendations and repair plans IAW CFCD 133 RCN Emergency Response Team Manual.

    4. Marking. This section is marked by a single assessor. The assessor is an officer delegated by the Convening Authority. The candidate is marked on 60% Conduct of ERT functions and 40% Communication based on guidelines identified in CFCD 133. PASS constitutes a minimum weighted average mark of 60%

5. **Time**. This exercise should last no more than 60 minutes.

6. **Remarks.** The Convening Authority is responsible for assessing this exercise. It is highly recommended that the assessment of this EO be conducted in the presence of SEA TRAINING GROUP (STG) staff under realistic conditions onboard the candidate's ship.

7. The candidate can complete this exercise as often as required at the discretion of

the Conveying Authority. Furthermore, the candidate can challenge this part of the board at any stage of the candidate's OJT period provided the candidate has completed the OJPR requirements of EO 012.02 first.

8. To achieve the highest levels of realistic assessment, it is recommended to include at least W Eng Tech Managers, the Snr Electrician, the MSEO (or designated officer), and the XO (or designated officer) in the conduct of the exercise.

9. Successful conduct of the exercise must be based only on the candidate's performance regardless of the strengths or weaknesses of the rest of the exercise participants. The Convening Authority scores the exercise based on direct observations and STG feedback (if present). Successfully challenging this exercise is prerequisite to challenging Part 2 of the board.

PART 2 – ORAL EXAMINATION BOARD

The HOD board is a technical and professional board. Therefore, the board's conduct shall be the pinnacle of professionalism.

Proper uniform, professional conduct, accurate terminology, and personal confidence are all vital to the candidate's success.

Commonly, the first two questions are completed (each of which must be individually passed) at which time the board will take a break to discuss carrying on. If the candidate is performing acceptably, the board will continue. If the candidate has failed either of the first two questions, the board may be ended and a debrief given. Actions upon failure of a board are IAW NAVORD 4500-20.

The board consists of 3 questions total, with the following recommended schedule and timings,

Q	Portion	Time (mins)	Morning Board 0800 Start	Afternoon Board 1300 Start
	Candidate presentation preparation	30	0730-0800	1230-1300
	Candidate and Board Member Introduction	10	0800-0810	1300-1310
1	Technical Failure	60	0810-0910	1310-1410
2	Technical Administration & Policy	25	0910-0935	1410-1435
	Break	20	0935-0955	1435-1455
3	Divisional Administration & Leadership	25	0955-1020	1455-1520
	Board Deliberation / Marking	15	1020-1035	1520-1535
	Initial Feedback and Completion	5	1035-1040	1535-1540

Question 1 | Technical Failure

30min to prepare. 60 minutes to present, including brief and questions.

A brief to the Commanding Officer regarding a failure scenario and the courses of action (CoAs) available, including an analysis of each one and the recommended CoA. Candidate is to brief on the incident, the initial actions taken and the operational impact of the failure. Use of the risk assessment process IAW the relevant NAVORDS and other references is expected. Candidates are to illustrate their follow-on actions and what organizations they will engage to assist or

provide support. The brief should include a risk assessment of the CoAs in accordance with NAVORD 3001-1, with the recommended CoA being that for which the reduction in risk (materiel, personnel and environmental) is as low as reasonably practicable (ALARP) in consideration of the resulting mission capability. The candidate shall note all reasonable assumptions made during the CoA development. The brief shall include follow on steps and a plan to implement and support the selected CoA including support and administration from shore facilities and other RCN departments.

Question 2 | Technical Administration & Policy

5 minutes preparation, 15 minutes delivery, 5 minutes for follow up questions.

A technical administration, policy, or operations question. This question can be one of any number of questions focusing on technical policy (e.g. work periods, engineering changes, etc.) or other form of policy such as docking procedures or technical management. The candidate shall have a detailed knowledge of the content and application of relevant policies and orders as well as an ability to discuss administrative resources and requirements from shore establishments. The Fleet Maintenance Facility, Formation Technical Authority, Fleet Engineering Operations, Fleet Engineering Readiness, Maritime Equipment Program Management, Ship-borne Helicopter Operating Procedures, Ship's Standing Orders, Naval Engineering Manual, Naval Materiel Management System, Cyber Mission Assurance and Cyber Material Assurance Programs, Technical Bulletins, and Technical Advisories are all within the scope of this question.

Questions can also include weapons certification, ammunition management, environment and safety regulations, and laser and radiation safety.

Question 3 | Divisional Administration and Leadership

5 minutes preparation, 15 minutes delivery, 5 minutes for follow up questions.

This question does not duplicate Leadership Respect Honour material, but is concerned with training progression for departmental personnel, disciplinary and administrative policy, and personnel management within a department. Candidates are to be given time to compose an answer. This question is used to assess the candidate's readiness to assume the role of Head of Department on a warship beyond technical policy. A strong awareness of corporate culture, ethos, and deck plate leadership is required. Question three focuses on the application of policy and the candidate's ability to not only discuss applicable policy, but also how to interpret and apply the policies in a situationally appropriate manner.

NCS Eng HOD Qualification Board Marking Guide – Part 1

Candidate:	**Ship:** HMCS
Qualification: AISF – Part 1	**Date:**
Convening Authority (or delegated officer):	**HOD:**
Sea Training Representative (if available):	

Instructions:
Part 1 of the HOD board is scored on this form. The candidate is marked on 60% Conduct of ERT functions and 40% Communication. PASS constitutes a minimum mark of 60%. Only one assessing officer is required to score and invigilate this part.

Score	**Conducting ERT**	Score	**Communication**
0	Candidate did not attempt to correlate any faults and the ERT organization completely disregarded Command Priorities.	0	Candidate briefs to Command and direction to ERT was not clear, concise, logical, confident or accurate.
1	Candidate failed to correlate any faults and the ERT organization was completely ineffective in meeting Command Priorities.	1	Candidate briefs to Command and direction to ERT was: (1 or 2 of clear, concise, logical, confident, and accurate).
2	Candidate struggled to correlate any faults and the ERT organization was generally ineffective in meeting Command Priorities.	2	Candidate briefs to Command and direction to ERT was: (3 of clear, concise, logical, confident, and accurate).
3	Candidate had difficulty correlating some faults and the ERT organization only partially supported Command Priorities.	3	Candidate briefs to Command and direction to ERT was: (4 of clear, concise, logical, confident, and accurate).
4	Candidate correlated some or all faults and the ERT organization supported Command Priorities.	4	Candidate briefs to Command and direction to ERT was clear, concise, logical, confident, and accurate.
5	Candidate skillfully correlated all faults, provided repair plans, and the ERT organization was highly effective in meeting Command Priorities.		
6	Candidate expertly correlated all faults, prioritized and executed repair plans, and the ERT organization proactively and effectively met Command Priorities.		

QUESTION	CONDUCTING ERT (0 – 6)	COMMS (0 – 4)	SCORE (S) (0 - 10)
Direct ERT			
Result: = Pass / Fail (Circle one)			
Comments:			
NOTE: Name and Signature are Mandatory **Name of Assessor**		**Signature**	

NCS Eng HOD Qualification Board Marking Guide – Part 2

Candidate:	**Ship:** HMCS	
Qualification: AISF – PART 2	**Date:**	
Chairperson:	**Examining Officer:**	**Command Representative:**
HOD:		

Instructions:
Part 2 results are scored on this form. Each question is scored for content and presentation which are weighted to create a score out of 10 for Q1 and Q2. The candidate shall achieve average mark of at least 60% for each scored question. Q1 and Q2 are weighted to create an overall board mark and the candidate must achieve a minimum weighted total mark of 60% to have successfully challenged Part 2.

Score	**Content**	Score	**Presentation**
0	Unable to answer any part of the question or even begin to discuss the topic.	0	Many major deficiencies
0.6	Unable to communicate a basic level of technical knowledge of engineering principles.	0.4	Some major deficiencies
1.2	Able to answer minor parts of the main question but unable to answer follow up questions.	0.8	Many intermediate deficiencies
1.8	Displays a very low level of technical knowledge of engineering principles.	1.2	Some intermediate deficiencies
2.4	Able to answer parts of the main question and a limited number of follow up questions displaying only a restricted level of technical knowledge.	1.6	Many minor deficiencies
3.0	Requires assistance answering the main question and requires prodding to answer follow up questions.	2.0	Some minor deficiencies
3.6	Displays an acceptable level of technical knowledge of engineering principles and administration requirements.	2.4	Acceptable with no major deficiencies in presentation
4.2	Displays an above average level of technical knowledge of engineering principles.	2.8	Good
4.8	Displays a high level of technical knowledge of engineering principles.	3.2	Very Good
5.4	Generates discussion through the display of excellent technical knowledge of engineering principles.	3.6	Excellent
6.0	Displays an exceptional level of technical knowledge answering the main question and follow up questions well beyond the expected level.	4.0	Exceptional

QUESTION	CONTENT (0 – 6)	PRESENTATION (0 – 4)	SCORE (S) (0 - 10)	WEIGHT FACTOR (WF)	WEIGHTED SCORE (WS = WF x S)
1. **Technical Failure**				6	
2. Technical Admin/Policy				4	
3. Divisional Admin Leadership				4	
TOTALS:				14	

Mark (%): = TWS x 10 = TWF

Result: = Pass / Fail (Circle one)

Comments:

NOTE: Name and Signature are Mandatory

Name of Board Member **Signature**

ANNEX J – ON JOB PERFORMANCE RECORD (OJPR)

(Certificate of Qualification)

Version	Revision Date	Remarks
OJPR Version 2.0	31 / 05 / 2023	OJPR Formatted into new RCN QSP format, original OJPR from 2011

Name and Initials	Rank	Service Number

Date Training Started	
Date of Completion	

THE OJPR MUST BE COMPLETED IN ACCORDANCE WITH THE INSTRUCTIONS BELOW. INCORRECT OR INCOMPLETE OJPR WILL BE RETURNED FOR RE-SUBMISSION.

INSTRUCTIONS

1. One copy of this OJPR, hereinafter referred to as the master copy, shall be made up for each individual beginning OJT. It shall be kept and maintained by the supervisor. The Training Establishment (TE) shall make a duplicate copy for retention by the trainee for the purpose of recording and reviewing individual progress. The TE for the OJPR must maintain Training Tracker fields as outlined in Monitor Mass.

2. As the trainee completes each Performance or Enabling Check to the required standard, IAW the associated checklist in ANNEX I, the Training PO or Officer will date and sign the applicable item on the master and duplicate copies of the OJPR sheet and inform the individual of the action taken.

3. It may not be possible to conduct OJT for some POs if, for example, the necessary equipment is not fitted on a ship, is not on issue to the unit, or is otherwise unavailable. If the training PO or Officer encounters any of these issues they are directed to contact Standards at the appropriate TE for guidance.

4. If an individual demonstrates the ability to perform additional duties not specified in this QSP, an appropriate entry shall be made under "Additional Duties" at the end of the OJPR.

PROGRESS REVIEW

5. Progress review shall be carried out quarterly by the Divisional Officer and the Training Officer.

PO/EO	Description	Remarks	Rank/Name/ Signature	Date	OPI
PO 001 Manage Departmental Personnel					
EO 001.01	Advise on Operational Limitations due to Personnel Shortages	Complete para 3a requirements			CSEO
		Complete para 3b requirements			XO
EO 001.02	Manage Individual Training and Education	Written report for 3a			CSEO
		Complete para 3b through 3e requirements			CSEO
EO 001.03	Provide Divisional support for Military Personnel	Complete Para 3 requirements			XO
		Written RW and C&P			CSEO
		Conduct briefings on para 3c			CSEO
EO 001.04	Manage Operation Training	Written Report for Para 3a			CSEO
		Complete Para 3b requirements			CSEO
PO 002 Liaise with Scientific establishments and Industry					
EO 002.01	Liaise with Scientific establishments and Industry	Written Report for Para 3			CSEO
PO 003 Manage Departmental Financial Resources					
EO 003.01	Manage Departmental Financial Resources	Complete Para 3a			CSEO
		Written report for para 3b through 3g			LOGO
PO 004 Manage Shipboard Risk					
EO 004.01	Promote environmental awareness onboard ship	Complete para 3			MSEO
EO 004.02	Manage interdaptmental safety procedures	Complete para 3a			UGSO
		Complete para 3b			XO
EO 004.03	Perform as Unit RF Safety Officer	Complete para 3a (Course)			CSEO
		Complete para 3b through 3d			CSEO
		Written report for 3e through 3g			CSEO

PO/EO	Description	Remarks	Rank/Name/ Signature	Date	OPI
EO 004.04	Perform the duties of the head of the HAZMAT cleanup team	Complete para 3			CSEO
EO 004.05	Ensure compliance with policies/procedures governing handling, stowage, and disposal of hazardous materials	Written report for para 3			CSEO
PO 005 Prepare Technical Documentation					
EO 005.01	Complete routine engineering reports	Complete para 3			CSEO
EO 005.02	Review and maintain technical directives, orders and records	Complete para 3			CSEO
EO 005.03	Prepare maritime technical policy	Complete para 3			CSEO
PO 006 Paticipate in Interdepartmental Operation Planning					
EO 006.01	Pariticipate in planning ship's operations, program, and/or routine	Briefings on para 3			CSEO
EO 006.02	Liaise with MSEO to assess impact of marine syste,s on combat systems	Written report for Para 3			MSEO
		Written report for Para 3			CSEO
PO 007 Manage Signature Management Program					
EO 007.01	Perform duties of Ship's Signature Management Officer	Complete para 3			CSEO
PO 008 Manage Shipbard configuration					
PO 008.01	Control configuration management onboard ship	Complete para 3			CSEO
PO 009 Manage Test and Trials Programme					
PO 009.01	Manage ship's test and trials programme	Complete para 3			CSEO
		Written report for para 3			CSEO
		Briefings on Para 3			CSEO
PO 010 Manage Readiness					
EO 010.01	Maximize combat systems availability	Complete para 3			CSEO

PO/EO	Description	Remarks	Rank/Name/ Signature	Date	OPI
	withing the stated notice for power	Written report para 3a (2) to (4), and 3a (9)			CSEO
		Briefings on para 3			CSEO
		Briefings on para 3			MSEO
		Brief the CO 10 times			CSEO
EO 010.02	Maximize combat systems availability at sea	Complete para 3			CSEO
		Written report para 3a (2) through (4)			CSEO
		Briefings on para 3			CSEO
		Briefings on para 3			MSEO
		Brief the CO 15 times			CSEO
EO 010.03	Advise on Ship's states of preparedness	Written report for para 3			CSEO
EO 010.04	Advise on shipboard degrees of readiness	Written report for para 3			CSEO
PO 011 Manage Defect Rectification					
EO 011.01	Direct and Supervise repair action due to equipment ailures	Complete para 3			CESO
		Written report for para 3c			LOGO
		Briefings on para 3			CSEO
		Briefings on para 3			OPSO
PO 012 Manage Battle Damage					
EO 012.01	Descript the combat systems emergency repair team organization	Written report for para 3			CSEO
EO 012.02	Direct Ship's ERT	Complete para 3			CSEO
		Act as CESO during Action Stations in ERTHQ (surface) or HQ 1 (sub) 3 times			CSEO /XO (subs)

PO/EO	Description	Remarks	Rank/Name/ Signature	Date	OPI
		Briefings on directing the ERT			CSEO
		Briefings on directing the ERT			OPSO / XO (subs)
EO 012.03	Manage ERT Training	Complete para 3			CSEO
		Complete Battle Problem			CSEO
					MSEO
					OPSO
PO 013 Respond to non-routine operations					
EO 013.01	Direct a HAZMAT clean-up team	Complete para 3 for a HAZMAT exercise			CSEO
EO 013.02	Response to ammunition emergencies	Written report for para 3			CSEO
		Briefings on various ammo emergencies			CSEO
PO 014 Manage Docking					
EO 014.01	Prepare ship for docking/undocking	Written report for para 3			CSEO
EO 014.02	Manage departmental actiosn while docked	Written report for para 3			CSEO
PO 015 Manage Ship's Tiered readiness program (TRP)					
EO 015.01	Project manage departmental responsibilities during a TRP	Written report for para 3			CSEO
		Briefings on TRP responsibilities			CSEO
					OPSO
PO 016 Manage formation level maintenance activities					
EO 016.01	Prepare and staff list line O&M documentation	Complete para 3			CSEO
EO 016.02	Liaise with formation staff wrt priority	Complete para 3			CSEO
		Briefs on various scenarios			CSEO
EO 016.03	Monitor contract work	Briefs on various scenarios			CSEO

PO/EO	Description	Remarks	Rank/Name/ Signature	Date	OPI
EO 016.04	Manage test equipment maintenance management information system (TEMMIS)	Complete para 3			CSEO
PO 017 Manage a short work period					
EO 017.01	Project manage a SWP	Complete para 3			CSEO
		Written report para 3			CSEO
		Act as CSEO during SWP			CSEO
		Briefings on managing a SWP			CSEO
PO 018 Manage ADM(Mat) Level maintenance activities					
EO 018.01	Plan a refit/DWP	Complete para 3			CSEO
		Written report para 3			CSEO
		Briefings on managing a refit/DWP			CSEO
EO 018.02	Monitor contract work	Complete para 3			CSEO
		Written report para 3			CSEO
EO 018.03	Provide support to QAR	Complete para 3			CSEO
		Written report para 3			CSEO
PO 019 Manage shipboard ammunition and explosive safety					
EO 019.01	Describe the ammunition organizations	Written report para 3			CSEO
		Briefing on ship and shore ammunition organizations			CSEO
EO 019.02	Manage the storage and handling of ammunition	Complete para 3			CSEO
		Written report para 3			CSEO
EO 019.03	Perform as ship's ammunition officer	Complete para 3			CSEO

PO/EO	Description	Remarks	Rank/Name/ Signature	Date	OPI
		Written report para 3			CSEO
		Complete UESO course			CSEO
EO 019.04	Manage the shipboard explosive safety program	Written report para 3			CSEO
PO 020 Manage departmental systems					
Eo 020.01	Describe above water sensor systems	Complete para 3			CSEO
EO 020.02	Describe communiciations systems	Complete para 3			CSEO
EO 020.03	Describe command and control systems	Complete para 3			CSEO
EO 020.04	Describe underwater sensor systems	Complete para 3			CSEO
EO 020.05	Describe eletrconic warfare systems	Complete para 3			CSEO
EO 020.06	Describe data processing systems	Complete para 3			CSEO
EO 020.07	Describe navigation systems	Complete para 3			CSEO
EO 020.08	Describe weapon systems	Complete para 3			CSEO
EO 020.09	Perform the duties of the CSEO	Briefings – equipment casualty scenarions			CSEO
		Act as the CSEO			CSEO
					CO

Progress Review Record

Authority & Period	Reviewed By	Signature	Date
Divisional Officer (Q1)			
Training Officer (Q1)			
Divisional Officer (Q2)			
Training Officer (Q2)			
Divisional Officer (Q3)			
Training Officer (Q3)			
Divisional Officer (Q4)			
Training Officer (Q4)			

TRAINING COMPLETION CERTIFICATE

I certify that_____has completed training to the standards stated in QSP – NCS Eng - AHOD, except as noted, and is recommended to challenge the Head of Department Qualification Board.

Date training completed:	
CSEO	Date
XO	Date
CO	Date

CERTIFICATION OF QUALIFICATION

I certify that _____ has completed the OJT to the standards stated in the Naval Combat Systems Engineering Assistant Head of Department QSP..

TE Cmdt/CO	Date

ANNEX K – TT PLAN

Nil TT plan due to being an OJPR QSP.

ANNEX L – RECORD OF DECISIONS TPWB

1. The original TPWB was sat during the QSWB, which the record can be found in annex D. It has not been duplicated in this annex.

RECORD OF AMENDMENT & AMPLIFICATION

1. The record of amendment and amplification will be filled in whenever a change to the TP is initiated. This section is used to provide amplification details and provide a history of the changes made. In the requirement for the amendment section, the member inputting the changes will transpose the information from the submitted TP Change Request (TPCR), thus ensuring the QSP has the most relevant information.

Version:	**3.1**
Date: 25 May 23	**Who Initiated Change:** HMCS Venture
Requirement for the Amendment: Update to new RCN QSP format, and update to all boards for NCS Eng and MS Eng	
Description of Changes: QSP was formatted into new format, no content changed from the previous QSP**.**	
Change Tracking number:	
Trg Mgr: Lt(N) Newcombe	

Previous course history

19 Mar 2018: NTDC(P) CO authorized update to Annex G, H, I that updated the purpose, structure, and execution of the PC 001. This update separated the board into Part 1: ERT, and Part 2: Oral Board.
27 Mar 2020: Amendment to Course Completion page (p. E-5 of QSP requiring signatures recommending procession to final board from the HoD, XO, and CO.
1 Apr 2020: Amendment to EO 004.03 Perform as RADHAZO (p. 4-18 of QSP) replacing all mentions of “RADHAZO” with “Unit RF Safety Officer”. All subsequent mentions throughout QSP amended as well.
6 Apr 2020: In Annex G PC 001 Qualification Board (p. G-1 of the QSP, in addition to the system casualty, divisional, and OP HONOUR questions, insertion of a fourth question dealing with technical issues and/or administration (e.g. SWP, TRP, ECs, support organizations, etc). The question shall have a weight of 4.
13 Jul 2020: Updated Foreword to reflect current organizations and authorities. Approving authority for Major Amendments is CNPTG, while AA for Minor Amendments is NTDC CO.
13 Jul 2020: Replaced “COMMANDANT OF CFNES” with “COMMANDER OF NPTG” on all Chapter title pages, as CFNES no longer exists.
13 Jul 2020: Amendment to Chapter 1 para 3, replacing “MARCORD 9-28” with “NAVORD 4500-20”. All MARCORDS to be superseded by equivalent NAVORDS.
13 Jul 2020: Replaced Chapter 1 para 11 with “This training will be validated IAW CFITES Vol. 8 – Validation of Instructional Programs, and based on the Task List in Annex B.” CFITES supersedes Naval Training Instructions.
13 Jul 2020: Replaced all mentions of “MARCORD 9-20” with “NAVORD 4500-20”.

13 Jul 2020: Replaced “MARCORD 9-60” with “NAVORD 9-60”. (p.3-1 para 5)
13 Jul 2020: Replaced “CFNES SO 425” with “NPTORD 4500-54. (p.3-2 para 6). CFNES no longer the authority.
13 Jul 2020: PO 005 Prepare Technical Documentation, replaced “MARCORDS” with “NAVORDS”. (p.4-28)
13 Jul 2020: Replaced all instances of “CBTO” with “OPSO”. Updated Annex F to remove CBTO and add OPSO (p.F-1)
13 Jul 2020: PO 013 Respond to non-routine operations, replaced “Vertical Launch/VLSS” with “GMVLS”. Removed para 3(d): “Fire in VLS fan room.” (p.4-60). VLS Fan Room was specific to IRO Class, all of which ships have been decommissioned.
13 Jul 2020: EO 019.01 Describe the Ammunition Organizations, para 3(b), removed DGMEPM (not involved with A&E) and DAPM (no longer exists), and inserted DAER, DAEME, and J4 Ammo. (p.4-84) Updated Annex F.
13 Jul 2020: EO 019.03 Perform as Ship’s Ammunition Officer, replaced “Operational Stock” and “OPSTOCK” with “Basic Load”, and replaced “Training Allocation” with “Baseline Ammunition.” Nomenclature change IAW NAVORD 7490.0. Replaced “Combat Team” with “Operations Team”. (p.4-89)
13 Jul 2020: Annex I NTO Qualification Board Marking Guide – Part 2, amended the scoring sheet (p.I-2) to include Question 4: Tech/Admin
28 Jul 2020: Foreword: Inserted explanatory phrase regarding the division of authorities within the Naval Training System.
28 Jul 2020: PO 007 replaced “MARCORD 46-109” with “NAVORD 3395-1”.
28 Jul 2020: Annex E replaced “MARCORD 9-20” with “NAVORD 4500-20” and CFNES/CFFSE with NPTG HQ.
25 May 2022: Replaced NAVORD 9-60 with NPTORD 4500-54 (Training Review Board Process)
8 Jul 22: Added UESO course as a requirement for PO019
21 Jul 22: Replaced NTDC(P) with HMCS VENTURE as CCA
Updated all refs of ‘SYO’ to ‘LOGO’
Updated all refs to ‘N37’ with ‘Fleet Engineering’ IAW org changes;
EO 001.04 Updated Assisting Officer to Assisting Member, IAW MJUL changes;
EO 004.02 added ‘Completing LOTO Audits’ to list of interdepartmental safety procedures
EO 004.03 added ‘Completing periodic RF Safety Audits as required by SEMS’
EO 005.03 added Ships’ SOPs, COTMs & fixed spelling of ‘Supersession’
EO 009.01 updated ‘CSAT’ to ‘CSAV’ (new terminology post-FELEX);
EO 013.03 made ‘Torpedo Hot Run’ generic instead of specific to Mk48, to account for new & different torpedo types (e.g. Mk54);
EO 016.01 replaced ‘MAFs and rewrote as ‘Reviewing and approving routine Stores demands’;
EO 001.01 Added SUBSAFE implications;
EO 001.04 updated to include Weapons Certifications; Submarine Weapon System Team Training; and Submarine Weapons Handling & Discharge Teams;
EO 004.04 Augmented to include submarine HAZMAT cleaning team & CSEO actions;
EO 005.02 augmented to include submarine-specific technical documentation & directives (SUBSAFE, certificates, registries, etc);
EO 005.03 augmented to include Submarine-specific documentation & policy (SMTI, VSTI, BRF, etc);
EO 006.01 added CASEX and TORPEX as examples of interdepartmental activities;
EO 009.01 added submarine-specific tests and trials (Camber Dive, Surface Trials, Dived Trials, Deep Dive, SONAR, Weapons and Ranging trials);
EO 010.03 Added relevant submarine states of readiness (Weapons & handling states; Signal Ejector states);

EO 011.01 Added submarine-specific repair considerations, including: Escape items; First level systems; License implications; and sub-specific DRMIS status meanings;
EO 012.01 Added requirement for officers in submarine to describe submarine damage control organization;
EO 012.02 Added requirements for officers in submarines to act as the CSEO in HQ1 and perform associated roles;
EO 012.03 Added training considerations for submarine DC organization (SCT, MCT, IPTT, etc)
EO 013.01 Added HAZMAT clean-up considerations specific to subs; mainly relating to Otto fuel cleanup IAW sub SOPs;
EO 013.02 Added submarine specific ammunition emergencies
EO 014.01 Added submarine specific docking & undocking requirements
EO 015.01 Added submarine specific TRP considerations, trials, etc.
EO 017.01 Added submarine specific SWP considerations (STO, DRMIS status, Risk Review Boards, VIC Maintenance cycle, etc)
EO 019.03 Added requirement to complete required explosive safety training (currently UESO course on DLN);

Added an initial list of useful References in Annex A, including general, surface and sub-surface sources. List has been left fairly generic, as a starting place for students' searches.

Removed empty table from Task List (Annex B) and provided a link to the Occupation Standard

Updated Training Completion certificate with clearer language and signature spaces recommending the candidate for the final board.

Annex E OJPR – Added signature block for HOD Board;

signature block for UESO course;

XO as signing authority for submarine CSEOs' DC/HQ1 Reqs;

Annex G Qualification Board – updated timing to 30 minutes;

removed LRH question and replaced with Technical Admin question (scoring weight of 4);

Added Subsafe implications to all 3 board questions for submariners;

cleaned up and clarified some wording

ANNEX M – ACRONYMS AND ABBREVIATIONS

1. This annex is to be utilized only to identify uncommon abbreviations used within this QSP.

Acyronm	
AA	Anti-Air
AAW	Anti-Air Warfare
AWW	Above Water Warfare
ACM	Acoustic Countermeasures
ADM(Mat)	Assistant Deputy Minister (Material)
AER	Automated Establishment Report
ARPS	Acoustic Range Prediction System
ASE	Automotive Service Excellence Standards
ASME	American Society of Mechanical Engineers
ASTN	Switched Transport Network Standards
ASuW	Anti-Surface Warfare
ASW	Anti-Submarine Warfare
ASWEX	Anti-Submarine Warfare Exercise
ATE	Automated Test Equipment
BAAL	Basic Ammunition Allocation List
BITE	Built-In Test Equipment
C&P	Counselling and Probation
CCP	Contract Change Proposal
CCS	Command and Control System
CEPA	Canadian Environmental Protection Act
CFAD	Canadian Forces Ammunition Depot
CFAO	Canadian Forces Administrative Orders
CFM	Contractor Furnished Material
CFQAR	Canadian Forces Quality Assurance Representative
CFSS	Canadian Forces Supply System
CFTO	Canadian Forces Technical Order
CHOD	Chief of Department
CIWS	Close-In Weapons Systems
CM	Configuration Management
CM	Corrective Maintenance
CMIS-S	Computerized Maintenance Information System – Ship's
CRR	Combat Readiness Report
CSAT	Combat System Alignment Test
CSE	Combat Systems Engineering
CSEOTI	Combat System Engineering Officer Technical Instruction
CSEOTM	Combat System Engineering Officer Temporary Memorandum
CTWSC	Combat Team Weapons Safety Certification
CWI	Continuous Wave Illuminator
DAEME	Directorate of Ammunition and Explosives Management and Engineering
DAER	Directorate of Ammunition and Explosives Regulation
DAOD	Defence Administrative Orders and Directives

DC	Damage Control
DCHQ	Damage Control Headquarters
DCOTT	Damage Control Organization Team Training
DDL	Dynamic Deliverable List
DMS	Defence Management System
DoDS	Department of Defence Standards
DRDC	Defence Research and Development Canada
DRE	Defence Research Establishment
DRP	Defect Rectification Period
DWP	Docking Work Period
EC	Engineering Change
ECP	Engineering Change Proposal
EDWP	Extended Docking Work Period
EHM	Equipment Health Monitoring
ELD	Equipment Logistic Directives
EMC	Electromagnetic Compatibility
EMC	Expenditure Management Course
EMI	Electromagnetic Interference
EMP	Electromagnetic Pulse
EMREQ	Emergency Requirements
EMS	Environmental Management System
EOOW	Engineering Officer of the Watch
EOTI	Engineering Officer Technical Instruction
EPM	Environmental Protection Management
ERT	Emergency Repair Team
ESL	Equipment Support List
Eval	Evaluation
EW	Electronic Warfare
EWP	Extended Work Period
EXDIR	Exercise Directive
FAA	Financial Administration Act
FAA	Formation Ammunition Authority
FDU	Fleet Diving Unit
FEL	Fitted Equipment List
FSR	Field Service Representative
FTA	Formation Technical Authority
FTB	Fleet Technical Bulletin
GMVLS	Guided Missile Vertical Launch System
GSM	Government Supplied Material
HAZMAT	Hazardous Material
HIFR	Helo In Flight Re-fuelling
HOD	Head of Department
HPR	High Priority Requisition
HVAC	Heating, Ventilation, and Air Conditioning
IEEE	Institute of Electrical and Electronics Engineer
IFF	Identify Friend or Foe
ILS	Integrated Logistic Support
IPT	In-Port Training
IR	Infrared

ISO	International Standards Organization
IT&E	Individual Training and Education
LCC	Life Cycle Costing
LCMM	Life Cycle Material Manager
LSA	Logistic Support Analysis
MAAL	MARLANT Ammunition Allowance List
MAF	Maintenance Action Form
(L)MRS	(Local) Maintenance Repair Specification
MARCORD	Maritime Command Order
MARLANTORD	Maritime Forces Atlantic Order
MARPACORD	Maritime Forces Pacific Order
MARPOL	International Convention for the Prevention of Pollution from Ships
MASIS	Material Acquisition and Support Information System
MILSTDS	Military Standards
MMIs	Material Management Instructions
MPAAL	MARPAC Ammunition Allowance List
MRC	Missile Readiness Checks
MRSL	Maintenance and Repair Specification List
MSE	Marine System Engineering
NaMMS	Naval Maintenance Management System
NAVORD	Naval Order
NBC	Nuclear, Biological, Chemical
NDE	Non-Destructive Examination
NDT	Non-Destructive Testing
NDID	National Defence Index and Documentation
NEI	Naval Equipment Index
NEM	Naval Engineering Manual
NES	National Exposure Standard
NETE	Naval Engineering Test Establishment
NFP	Notice for Power
NTDS	Naval Tactical Data System
OCCAP	Oil and Coolant Condition Analysis Program
OEM	Original Equipment Manufacture
OHSA	Occupational Health and Safety Act
OPDEF	Operational Deficiency
OPSCHED	Operational Schedule
OPSO	Operations Officer
ORO	Operations Room Officer
OT	Operational Training
OTT	Operational Team Training
PIF	Pre-Installation Failure Report
PL	Project Leader
PM	Preventative Maintenance
PM	Project Management/Manager
PMRS	Particularized Maintenance Repair Specification
PSL	Particularized Specification List
PWGSC	Public Works and Government Services Canada
QA	Quality Assurance
QAR	Quality Assurance Representative

QC	Quality Control
QETE	Quality Engineering Test Establishment
QHM	Queen's Harbour Master
QR&O	Queen's Regulations and Orders
R&D	Research and Development
R&O	Repair and Overhaul
R&SP	Readiness and Sustainment Policy
RADHAZ	Non-Ionizing Radiation Hazard
RAM	Reliability, Availability and Maintainability
RCM	Reliability Centered Maintenance
REMAR	Report on the Management of Resource
RF	Repair Facilities
RSP	Readiness Support Program
RTI	Request for Technical Information
RW	Recorded Warning
SAR	Search and Rescue
SBT	Section Base Team
SHC	Stock Holding Code
SMMISM	Ship's Maintenance Management Information System Manual
SOCD	Statement of Operational Capability Deficiency
SOP	Standard Operation Procedure
SOR	Statement of Requirements
SOT	System Operability Test
SOW	Statement of Work
SPATE	Special Tools and Test Equipment
SPDET	Safety Program Development and Evaluation Technique
SPF	SONAR Performance Figure
SRB	Senior Review Board
SRI	Sea Readiness Inspection
SS	Ship's Staff
SSE	Ship Systems Engineering
SSO	Ship's Standing Officers
SSRWC	Standard Ship Refit Work Catalogue
SWOAD	Ship Without Air Detachment
SWP	Short Work Period
TEMMIS	Test Equipment Maintenance Management Information System
TEWA	Threat Evaluation Weapons Assignment
TMA	Target Motion Analysis
TRANREQ	Transfer Requirement
TSOR	Technical Statement of Requirements
UCR	Unsatisfactory Condition Report
VA	Vibration Analysis
VSTI	Victoria Special Technical Instructions
WSAT	Weapons System Accuracy Trial
WRPS	Work Request Priority Schedule
WSB	Watch and Station Bill